

UNIVERSITY OF BUEA

P.O. BOX 63,

Buea, South West Region

CAMEROON

Tel: (237) 3332 21 34/3332 26 90

Fax: (237) 3332 22 72



REPUBLIC OF CAMEROON

PEACE-WORK-FATHERLAND

**FACULTY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER ENGINEERING**

**ANALYSIS, DESIGN, AND IMPLEMENTATION OF A
SCHOOL MANAGEMENT SYSTEM FOR FET**

A dissertation submitted to the Department of Computer Engineering, Faculty of Engineering and Technology, University of Buea, in Partial Fulfilment of the Requirements for the Award of Bachelor of Engineering (B.Eng.) Degree in Computer Engineering.

BY:

Takem Jim-Rawlings E.

Matriculation Number: FE21A309

Option: Software Engineering

Supervisor:

Dr. DJOUELA Ines

University of Buea

2024/2025 Academic Year

**ANALYSIS, DESIGN, AND IMPLEMENTATION OF A SCHOOL
MANAGEMENT SYSTEM FOR FET**

Takem Jim-Rawlings E.
Matriculation Number: FE21A309
2024/2025 Academic Year

*Dissertation submitted in partial fulfilment of the Requirements for the award of
Bachelor of Engineering (B.Eng.) Degree in Computer Engineering.*

**Department of Computer Engineering
Faculty of Engineering and Technology
University of Buea**

CERTIFICATION OF ORIGINALITY

We the undersigned, hereby certify that this dissertation entitled **ANALYSIS, DESIGN, AND IMPLEMENTATION OF A SCHOOL MANAGEMENT SYSTEM FOR FET** presented by **TAKEM JIM-RAWLINGS E.**, matriculation number **FE21A309** has been carried out by him in the Department of Computer Engineering, Faculty of Engineering and Technology, University of Buea under the supervision of **Dr. DJOUELA Ines**

This dissertation is authentic and represents the fruits of his own research and efforts.

Date: _____

Student:

Supervisor:

Head of Department:

DEDICATION

To my wonderful parents, Takem James Eyong and Takem Martha Ayuk, for their support, sacrifice, and endless love.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the Dean of the Faculty of Engineering and Technology, **Prof. Agbor Dieudonne Agbor**, the Head of Department, **Prof. Elie Fute**, and the entire faculty for providing me with the knowledge and resources to excel in my field of study.

A special thank you goes to my academic supervisor, **Dr. Ines Djouela**, for going the extra mile in providing guidance, feedback, and continuous encouragement throughout my academic journey and the writing of this report.

I am deeply thankful to my parents for their unconditional love, encouragement, and for sponsoring me throughout my education. Without their sacrifices and support, I would not have reached this milestone.

I also appreciate my brother, **Fidel**, for his constant love, encouragement, and support, as well as my wonderful friends for their encouragement and companionship throughout this journey.

Most especially, I thank **Almighty God** for His wisdom, strength, and good health that have sustained me throughout my studies. His blessings have been the foundation of my success.

Thank you all for your support and for being part of this significant chapter of my life.

ABSTRACT

The rapid integration of Information and Communication Technology (ICT) in education has necessitated efficient systems to manage academic and administrative processes. This study focuses on the analysis, design, and implementation of a School Management System (SMS) tailored for the Faculty of Engineering and Technology (FET) at the University of Buea, Cameroon. The existing reliance on informal communication platforms, such as WhatsApp, has led to fragmented information dissemination, missed updates, and inefficiencies in course management, negatively impacting student engagement and academic performance.

The objective of this project is to develop a centralized, user-friendly School Management System (SMS) to ease communication, course management, and administrative tasks for students, lecturers, and administrators. Utilizing the Agile development methodology, the system is designed with a multi-tier architecture, incorporating React.js and React Native for frontend interfaces, Node.js with Express.js for backend services, and PostgreSQL for data management. Key features include general and course-specific announcements, assignment management, and offline access, addressing FET's specific needs. Data was collected through surveys and interviews with stakeholders to identify requirements, followed by iterative development and testing. The resulting FET SPACE system offers role-based access, real-time updates, and a distraction-free interface, significantly improving academic coordination and resource accessibility compared to existing tools like Google Classroom and WhatsApp.

The system's context-specific design makes it a cost-effective solution for FET, with potential scalability to other faculties. Future enhancements include attendance tracking, timetable generation, and lecturer rating modules to further support FET's academic mission.

Keywords: School Management System, Learning Management System, Faculty of Engineering and Technology, Agile Methodology, Centralized Communication, Course Management, Academic Engagement, ICT in Education.

TABLE OF CONTENTS

CERTIFICATION OF ORIGINALITY	ii
DEDICATION	iii
ACKNOWLEDGEMENT	iv
ABSTRACT.....	v
TABLE OF CONTENTS.....	vi
LIST OF TABLES	x
LIST OF FIGURES	xi
LIST OF ABBREVIATIONS.....	xii
CHAPTER ONE: GENERAL INTRODUCTION	1
1.1. Background and Context of the Study	1
1.2. Problem Statement	1
1.3. Objectives of the Study	2
1.3.1. General Objective	2
1.3.2. Specific Objectives	2
1.4. Proposed Methodology	3
1.4.1. Data Collection	3
1.4.2. Development Process	3
1.5. Significance of the Study	4
1.5.1. Significance to Students	4
1.5.2. Significance to Lecturers	4
1.5.3. Significance to Administrators	4
1.5.4. Significance to Future Researchers and Developers	4
1.6. Scope of the Study.....	4
1.7. Delimitation of the Study	5
1.8. Definition of Keywords and Terms.....	6
1.9 Organization of the Dissertation	6

CHAPTER 2. LITERATURE REVIEW	8
2.1 Introduction	8
2.2 General Concepts on School Management Systems	8
2.2.1 School Management System (SMS) / School Management Information System.....	8
2.2.2 School Management System vs Learning Management System.....	9
2.2.3 Student Information System (SIS).....	9
2.2.4 Management Information Systems (MIS) in Education	10
2.2.5 Centralization of Data and Processes	10
2.2.6 Automation of Administrative Tasks	10
2.2.7 Enhancement of Communication and Collaboration.....	11
2.2.8 Support for Data-Driven Decision Making	11
2.2.9 Focus on System Quality and Implementation Factors	11
2.3 Related Works	11
2.3.1 Shah, M. (2014). Impact of Management Information Systems (MIS) on School Administration: What the Literature Says. <i>Procedia - Social and Behavioral Sciences</i> , 116, 2799–2804	12
2.3.2 Gehlawat, M. (2014). School Management Information System: An Effective Tool for Augmenting the School Practices. <i>New Frontiers in Education</i> , 47(2), 57–64.....	13
2.3.3 JayaSudha, A. R., & Arunachalam, C. P. (2023). School Management System. <i>Journal of Advance Research in Mobile Computing</i> , 5(1), 35–39.....	14
2.3.4 Adesina, A. O., Ajagbe, S. A., Odule, T. J., & Agbele, K. K. (2022). Development of an Improved School Information Management System. <i>FUW Trends in Science & Technology Journal</i> , 7(1), 120–134	15
2.3.5 Orhani, S., Saramati, E., & Drini, L. (2023). School Administration Through the School’s Electronic Management System. <i>European Journal of Educational Management</i> , 6(1), 59–67	17
2.3.6 Grepon, B., Baran, N., Gumonan, K., Martinez, A., & Lacsa, M. (in press). Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines	18

2.3.7 Caratiquit, K. D. (2021). Web-based School Information and Publication System: A Developmental Study	20
2.4 Partial Conclusion	21
CHAPTER 3: ANALYSIS AND DESIGN	23
3.1. Introduction	23
3.2. Methodology	23
3.3. Requirements Gathering and Analysis	23
3.3.1. Requirements Gathering	23
3.3.2. Requirements Analysis	25
Functional and Non-Functional Requirements	27
Prioritized Requirements	28
3.4 Design.....	29
3.4.1 System Modeling.....	29
3.4.1.1. Use Case Diagram	29
3.4.1.2. Class Diagram.....	32
3.4.1.3. Sequence Diagram	35
3.4.1.6. Entity Relationship Diagram	39
3.4.2 UI Design.....	42
3.5. Global Architecture of the Solution	44
3.5.1 Client Layer	44
3.5.2 Application Layer	45
3.5.3 Data Layer	45
3.6. Partial Conclusion	46
CHAPTER 4. IMPLEMENTATION AND RESULTS.....	47
4.1. Introduction	47
4.2. Tools and Materials Used.....	47
4.3. Description of the implementation process	48
4.3.1 Environment Setup	48

4.3.2 Frontend Development	48
4.3.3 Data Management with React Query	49
4.3.4 Backend Development.....	49
4.4. Presentation and Interpretation of Results	50
I. Student Panel.....	50
II. Teacher Panel.....	51
III. Admin Panel (Web View Only)	52
4.5. Evaluation of the Solution and Comparative Analysis	53
4.5.1. Comparative Analysis of Existing Systems.....	53
4.5.2. FET SPACE as a Contextual Solution	54
4.6 Deployment and Execution Instructions	55
4.7. Partial Conclusion	56
CHAPTER 5. CONCLUSION AND FURTHER WORKS	57
5.1. Summary of Findings	57
5.2. Contribution to Engineering and Technology	57
5.3. Recommendations	58
5.4. Difficulties Encountered	58
5.5. Further Works	58
5.6. Partial Conclusion	59
REFERENCES	60

LIST OF TABLES

Table 1: Summary of teacher requirements gotten after interview.....	24
Table 2: Summarized comparison of the school management system FET space and the existing solutions	54

LIST OF FIGURES

Figure 1: Student preferences for platform features considered essential for future development. ..	26
Figure 2: Pie chart showing the students preferred platform access methods	27
Figure 3: Use case diagram for the school management system	30
Figure 4: Class diagram for the school management system.....	32
Figure 5: User Onboarding Flow	36
Figure 6: Course Access Management	37
Figure 7: Assignment Handling	38
Figure 8: ER Diagram for the school management system	40
Figure 9: Wireframes of school management system.....	43
Figure 10: High fidelity prototype of school management system.....	44
Figure 11: System Architecture Overview for school management system.....	44

LIST OF ABBREVIATIONS

AI:	Artificial Intelligence
API:	Application Programming Interface
CI/CD:	Continuous Integration / Continuous Deployment
CRUD:	Create, Read, Update, Delete
CSS:	Cascading Style Sheet
ER:	Entity Relational
FET:	Faculty of Engineering and Technology
LMS:	Learning Management Syse
MVP:	Minimum Viable Product
ORM:	Object-Relational Mapping
SMS:	School Management System
SQL:	Structured Query Language
UI/UX:	User Interface/User Experience
UML:	Unified Modeling Language

CHAPTER ONE: GENERAL INTRODUCTION

1.1. Background and Context of the Study

Over the past decade, the integration of ICT in education has transformed how institutions operate, communicate, and deliver learning. Among the tools facilitating this transformation are SMSs (School Management Systems). These are digital platforms designed to centralize and ease both academic and administrative activities within educational institutions.

The FET (Faculty of Engineering and Technology) at the University of Buea, is a cornerstone of technical education in Cameroon, offering programs in electrical, computer, mechanical, and civil engineering, among others. With an estimated enrollment of over 500 students and a diverse faculty, FET delivers rigorous academic curricula to prepare students for industry demands. The faculty operates within a dynamic environment where effective communication and resource access are vital for academic success.

Traditionally, FET relies on informal communication channels, primarily WhatsApp group chats, to disseminate academic information, including lecture schedules, course materials, and administrative announcements. These platforms, while widely accessible, reflect the broader trend of leveraging technology in academic settings. As FET's enrollment grows and academic programs expand, the demand for streamlined communication and management systems has intensified. Students require timely access to course content, lecturers need efficient tools for sharing resources, and administrators seek reliable methods for coordinating faculty operations and passing out important information. This increases the need for a structured, technology-driven solution to enhance engagement and efficiency.

To address these challenges, this study explores the potential of a tailored SMS, focusing on improving information dissemination and academic coordination fostering FET's mission to deliver high quality technical education, all in a distraction-free environment.

1.2. Problem Statement

The current communication and course management system at FET relies heavily on WhatsApp group chats, which are inadequate for addressing the academic needs of students, lecturers, and administrators. Important academic information is typically shared in a single class group chat and then manually forwarded to other groups. Lecture notes, assignments, and announcements are shared informally across chats, often leading to fragmented and unreliable communication. This unstructured

method frequently results in missed updates, as not all students belong to every group or consistently receive forwarded messages.

Moreover, critical content is often buried under casual, non-academic conversations or lost in lengthy chat histories, making it difficult for students to locate and retrieve essential academic materials. This disorganization causes delays in accessing resources, confusion over academic requirements, and late submissions—factors that negatively affect student engagement and academic performance.

Students have also expressed challenges in managing academic content—such as notes, revision questions, and course-specific updates—due to information being scattered across multiple platforms or not consistently made available. The absence of a centralized repository introduces distractions, reduces academic focus, and limits overall productivity.

Lecturers face repeated inefficiencies in sharing learning resources, as platforms like WhatsApp lack proper archiving functionality and organizational capabilities. This often forces them to repost materials multiple times, especially whenever there is a change in batch, class, or academic level.

These challenges, observed consistently over recent academic years, hinder FET's ability to support effective teaching, learning, and academic planning. In a faculty managing a growing student population, the absence of a structured, centralized system compromises academic quality and operational coordination. There is an urgent need for a formal School Management System to streamline communication, ensure consistent access to academic resources, and enhance overall faculty efficiency.

1.3. Objectives of the Study

This section presents the overall aim of the study and breaks it down into more specific, measurable goals.

1.3.1. General Objective

To design and implement a centralized School Management System for FET Buea that facilitates communication and information dissemination, centralizes course management, and academic engagement for students, lecturers, and administrators.

1.3.2. Specific Objectives

- To investigate and document the current challenges in communication and course content

management within FET Buea, identifying key challenges and stakeholders (students, lecturers, and administrators) needs through surveys, interviews, and observations.

- To analyze and design a modular architecture that accommodates key academic features such as announcements, content uploads, and assignment submission.
- To design a distraction-free interface that minimizes non-academic interactions, thereby improving student focus and productivity.
- To develop user-specific frontend interfaces for students, lecturers, and administrators, ensuring intuitive design, usability, and accessibility tailored to each role.
- To design and develop a secure and scalable backend system that handles data management, authentication, authorization, and business logic using modern web technologies.
- To integrate the frontend interfaces with the backend services through RESTful APIs or other appropriate communication protocols, ensuring role-based access and smooth data exchange.
- To deploy a centralized digital platform that replaces fragmented communication methods (e.g., WhatsApp groups) with structured, institution-wide academic communication tools.
- To evaluate the usability, performance, and effectiveness of the system through testing and user feedback, ensuring it meets the academic and administrative needs of FET Buea.

1.4. Proposed Methodology

This section provides an overview of how the research and system development will be conducted. The system will be developed using the Agile Development Methodology, which will emphasize iterative development, stakeholder feedback, and flexibility in design. This approach will enable continuous user input and progressive system refinement over several development cycles.

1.4.1. Data Collection

To understand user needs and expectations, data will be collected from various stakeholder groups:

- Students will participate in both online surveys and in-person interviews to describe current pain points and propose desired features.
- Lecturers and administrators will provide insights through structured interviews, focusing on their challenges with communication and student engagement.

1.4.2. Development Process

The methodology will include the following stages:

- **Requirement Gathering:** Collecting needs from different user groups.
- **Requirement Analysis:** Categorizing and validating requirements.
- **System Design:** Creating architectural design, interface planning, and feature layout.
- **UI/UX Design:** Developing wireframes, prototypes, and visual interfaces.
- **Implementation:** Coding the system using relevant web/mobile technologies.
- **Testing and Feedback:** Conducting user testing, feedback sessions, and minor revisions

1.5. Significance of the Study

The development of a School Management System for FET Buea holds relevance for multiple stakeholders within the academic community. This section outlines its significance across various groups.

1.5.1. Significance to Students

The system provides students with a structured and centralized platform where they can access course materials, submit assignments, and receive relevant announcements. It enhances academic organization and reduces dependency on unstructured, informal communication platforms.

1.5.2. Significance to Lecturers

For lecturers, the system simplifies the distribution of educational materials, facilitates student communication, and provides a more organized method for managing academic tasks such as assignments and deadlines.

1.5.3. Significance to Administrators

The system offers administrators a formal channel to disseminate faculty-wide announcements, track user activity, and maintain oversight over academic communications in a centralized manner. It improves institutional control over communication workflows.

1.5.4. Significance to Future Researchers and Developers

This project serves as a useful reference for researchers and software developers interested in designing academic platforms for institutions with similar needs. It provides a real-world case study of implementing school management systems in a resource-limited context

1.6. Scope of the Study

This study focuses exclusively on the Faculty of Engineering and Technology (FET) at the University of Buea. It targets three primary user groups: **students, lecturers, and administrators**. The School Management System developed addresses key academic functionalities, namely:

- Structured announcement distribution
- Centralized access to course materials
- Assignment management (submission and feedback)

The system is designed specifically for the FET environment and does not extend to other faculties or departments within the university. Additionally, the study is limited to data collection, testing, and evaluation carried out during the 2024–2025 academic year. All user feedback and performance assessments are based solely on inputs from FET stakeholders.

This narrowly defined scope was deliberately chosen to ensure project feasibility within the allotted timeline and to facilitate focused, meaningful evaluation of the system’s effectiveness in a well-defined academic setting.

1.7. Delimitation of the Study

While the School Management System is designed to address core academic needs within FET, certain requirements have been intentionally deferred for future implementation. This decision was guided by practical constraints, including limited time, technical resources, and project scope.

The initial version of the system focuses on delivering essential features with the highest immediate impact—namely communication, course material access, and assignment management. The following additional features, though recognized as valuable, were excluded from this phase and are planned for future development:

- **Finance-related modules (e.g., fee payment):**
Deferred because financial operations are currently managed by existing systems within the university.
- **Timetable generation:**
Postponed due to the complexity of developing a dynamic scheduling engine that must coordinate multiple factors such as class venues, time slots, and course overlaps.
- **Attendance tracking:**
Set aside for future phases due to its dependency on real-time classroom integration and additional implementation complexity.

- **Automated questionnaire generation:**

Excluded initially due to its technical demands and relatively lower priority in addressing immediate academic workflow needs.

These components are acknowledged as important and will be considered in subsequent phases of the system, depending on institutional priorities, user feedback, and the availability of additional development resources.

1.8. Definition of Keywords and Terms

This section defines key terms used throughout the study to ensure clarity and shared understanding.

- **School Management System (SMS):** A digital platform that integrates administrative and academic functions such as announcements, schedules, academic records, and content delivery.
- **Learning Management System (LMS):** A component or standalone system focused on delivering, tracking, and managing educational content and learner activities. Often considered a subset of SMS.
- **Course Management:** The administration and organization of academic materials, assignments, and communications for a specific course.
- **Content Centralization:** The process of organizing all related academic materials in a single, accessible digital space.

1.9 Organization of the Dissertation

This dissertation is structured into five main chapters, each of which addresses a specific aspect of the research and system development process.

- **Chapter One: General Introduction**

This chapter provides the background to the study, defines the problem under investigation, outlines the study's objectives and scope, and introduces key definitions. It also describes the methodology and the structure of the dissertation.

- **Chapter Two: Literature Review**

This chapter examines existing literature related to School Management Systems and Learning Management Systems. It highlights the importance, benefits, and challenges of such systems, and explores their relevance in improving academic communication and course

management in higher education.

- **Chapter Three: Analysis and Design**

This chapter discusses the adopted agile development methodology and the data collection process. It also presents the analysis of system requirements, use case modeling, and the architectural design of the proposed School Management System.

- **Chapter Four: Implementation and Results**

This chapter describes the implementation of the system, the tools and technologies used, and the features developed. It presents an overview of each system page, explains the design decisions made, and discusses the results obtained following the development process.

- **Chapter Five: Conclusion and Future Work**

This chapter provides a summary of the findings and revisits the research objectives to assess how they were addressed. It also reflects on the system's limitations and proposes areas for future development, including the integration of features such as timetable generation, attendance tracking, and financial management.

This chapter has laid the groundwork for the study by presenting the background, context, and problem statement, alongside the objectives, methodology, significance, scope, delimitations, definitions, and structure of the dissertation. By defining the problem and outlining the study's goals, this chapter establishes a clear foundation for investigating a centralized SMS to support FET's mission. The subsequent chapters will build on this foundation, exploring existing literature, detailing the system's design and implementation, and evaluating its impact on FET's academic community.

CHAPTER 2. LITERATURE REVIEW

2.1 Introduction

The rapid advancement of information technology has transformed the administration of educational institutions, placing School Management Systems (SMS) at the forefront of this evolution. An SMS is a large computer system that provides schools with the ability to support day-to-day operations within the institution (JayaSudha & Arunachalam, 2023). These systems have become essential tools for managing student records, staff activities, and institutional operations, enhancing efficiency, transparency, and stakeholder engagement in schools worldwide (Shah, 2013; Writer, 2021). One sector that has significantly benefited from these advancements is the education sector, which remains one of the largest and most dynamic industries globally. As long as children continue to engage in learning, this sector will grow and thrive, incorporating continuous improvements and innovations into its processes (Writer, 2021). The use of information technology in educational management has rapidly increased due to its efficiency and effectiveness (Shah, 2013), offering a technological solution to streamline tasks such as attendance tracking, grade reporting, and resource allocation, that were once labor-intensive and error-prone (Orhani et al., 2023; Grepon et al., 2021). However, their adoption is not without challenges, including implementation costs, user training needs, and ensuring system scalability across diverse educational contexts (Adesina et al., 2022; JayaSudha & Arunachalam, 2023).

This chapter provides a comprehensive review of the existing literature on School Management Systems, situating the current study within the broader domain of educational technology research. It begins by exploring general concepts related to SMS, including their definitions, functionalities, and implications for school administration. The review then examines a range of tools, techniques, and methodologies employed in SMS development and implementation, drawing on key findings from related works to highlight their strengths and limitations. Finally, future directions, such as the integration of artificial intelligence and mobile technologies, are considered to address emerging needs in educational management.

2.2 General Concepts on School Management Systems

2.2.1 School Management System (SMS) / School Management Information System

A School Management System (SMS), often interchangeably termed a School Management Information System (SMIS), is a software platform designed to ease the administrative and academic

operations of educational institutions. It encompasses a wide range of functionalities, including student record management, attendance tracking, and resource allocation (Shah, 2014; Gehlawat, 2014). Studies like Orhani et al. (2023) and Grepon et al. (in press) illustrate SMSs as web-based or hybrid systems that centralize school processes, enhancing efficiency and stakeholder access. Online sources, such as EdTech Magazine (2023), emphasize SMSs as pivotal in modernizing education, integrating tools like dashboards and real-time updates to meet institutional needs.

2.2.2 School Management System vs Learning Management System

A **School Management System (SMS)** and a **Learning Management System (LMS)** are both digital platforms used in educational institutions, but they serve distinct purposes and functions. While an SMS focuses on managing institutional operations such as student records, staff information, and administrative logistics, an LMS primarily supports the teaching and learning process by facilitating content delivery, assessment, and student progress tracking (SchoolCues, 2023).

An SMS is typically used by administrators to oversee various structural aspects of a school, such as attendance tracking, timetable scheduling, finance and fee management, and academic performance recording (Shah, 2014; Gehlawat, 2014). Conversely, an LMS is designed to assist educators and learners, providing features such as assignment distribution, lecture uploads, interactive assessments, and real-time feedback (Orhani et al., 2023).

Despite their differing roles, both systems share several similarities. Each platform contributes to data and information management, provides tools for student engagement tracking, features user-friendly interfaces, and offers cloud-based or on-premise deployment models. Additionally, both enhance communication and collaboration between stakeholders, such as teachers, students, and parents (SchoolCues, 2023).

2.2.3 Student Information System (SIS)

A **Student Information System (SIS)** is a specialized subset of SMS focused on managing student-specific data, such as enrollment details, grades, and attendance records. Caratiquit (2021) highlights its role in providing downloadable resources and updates for students, while Adesina et al. (2022) emphasize SIS features like truancy monitoring and parental access to academic progress. Often integrated within broader SMS frameworks, SISs leverage databases (e.g., MySQL in JayaSudha &

Arunachalam, 2023) to ensure accurate, accessible student data, aligning with broader educational goals of transparency and accountability (UNESCO, 2022).

2.2.4 Management Information Systems (MIS) in Education

Management Information Systems (MIS) in education are technological frameworks that enhance administrative decision-making and operational efficiency. Shah (2013) defines MIS as systems supporting activities like attendance monitoring, financial management, and resource allocation, providing managers with information to manage effectively. O'Brien (1999) describes MIS as a discipline integrating computer systems with organizational goals, focusing on strategic and operational analysis. Waston et al. (1987) characterize it as a method delivering past, present, and projected data on internal operations and external intelligence to aid decision-making, while Telem (1999) specifies it as a system tailored to a school's structure and needs. Shah (2013) traces MIS evolution from clerical tools to strategic assets, and Gehlawat (2014) highlights modules like Staff Management enhancing governance. Forbes (2023) notes their role in integrating data streams, optimizing resources for administrators.

2.2.5 Centralization of Data and Processes

Centralization is a core SMS concept, consolidating data and processes into a unified platform for streamlined management. Grepon et al. (in press) describe a client-server architecture that centralizes enrollment and grading at Northern Bukidnon State College, reducing redundancies. Similarly, Orhani et al. (2023) highlight an online database at "Heronjtë e Lumës" for storing teacher and student data, improving accessibility. TeachThought (2022) underscores centralization's role in reducing data silos, ensuring consistency and enabling real-time updates across departments.

2.2.6 Automation of Administrative Tasks

Automation within SMSs eliminates manual processes, enhancing efficiency and accuracy. JayaSudha & Arunachalam (2023) demonstrate automation of fee payments and attendance in a PHP-based system, while Adesina et al. (2022) automate attendance and result computation to curb truancy. Gehlawat (2014) lists automated modules like Timetable and Examination Management, reducing teacher workload. TechRadar (2024) notes that automation minimizes human error and frees staff for strategic tasks, a benefit echoed across the reviewed studies.

2.2.7 Enhancement of Communication and Collaboration

SMSs enhance communication and collaboration by connecting stakeholders through digital channels. Caratiquit (2021) integrates social media and portals for sharing school updates, fostering community engagement at Lal-lo National High School. Orhani et al. (2023) report improved teacher-parent communication via their ESMS, with 91.7% of teachers noting facilitation benefits. eSchool News (2023) highlights SMSs as virtual hubs, enabling forums, notifications, and feedback loops that strengthen school-home partnerships.

2.2.8 Support for Data-Driven Decision Making

SMSs support data-driven decision-making by providing actionable insights through analytics and reporting. Shah (2014) emphasizes MIS's role in generating timely reports for planning, a feature Grepon et al. (in press) implement with enrollment and grade summaries for strategic use. Adesina et al. (2022) include a Student Evaluation of Teaching component to assess performance, enhancing accountability. Online sources like UNESCO (2022) stress that data-driven SMSs empower administrators to align resources with educational goals, improving outcomes.

2.2.9 Focus on System Quality and Implementation Factors

System quality and implementation factors are critical to SMS success, often evaluated via standards like ISO 25010. Caratiquit (2021) and Grepon et al. (in press) achieve "Excellent" and "Very Good" ratings in functionality, usability, and reliability, respectively, ensuring user satisfaction. Implementation challenges, such as training needs (Orhani et al., 2023) and infrastructure costs (Shah, 2014), highlight the importance of user readiness and technical support. EdTech Magazine (2023) advises that scalability, security, and maintainability are key considerations, requiring ongoing updates and stakeholder buy-in for sustained impact.

2.3 Related Works

This section reviews key studies that explore the role, implementation, and impact of School Management Systems (SMSs) and Management Information Systems (MIS) in educational administration. The works span diverse contexts, highlighting their importance in enhancing efficiency, their implementation methodologies, and key findings on stakeholder engagement and system efficacy.

Shah (2014) highlights the broad impact of MIS, detailing modules like examination, staff, timetable, student, and attendance management systems, emphasizing their strategic importance in decision-making. Gehlawat (2014) explores SMIS's role in augmenting school practices, focusing on its comprehensive data management capabilities. JayaSudha and Arunachalam (2023) demonstrate practical implementation using PHP CodeIgniter, prioritizing accurate student activity records and role-based access for stakeholders. Adesina et al. (2022) present a web-based SMS to curb truancy, implemented on a local server with distinct user privileges. Orhani et al. (2023) reveal positive teacher perceptions of electronic SMS in Kosovo, affirming its operational benefits. Grepon et al. (2021) evaluate an SMIS using the Agile model and ISO 25010 standards, achieving high functionality and usability. Caratiquit (2021) showcases a web-based system enhancing online information sharing, noted for cost savings and real-time access. These studies collectively highlight SMSs' transformative potential, implementation challenges, and stakeholder impacts, laying a foundation for understanding their significance in modern educational settings.

2.3.1 Shah, M. (2014). Impact of Management Information Systems (MIS) on School Administration: What the Literature Says. *Procedia - Social and Behavioral Sciences*, 116, 2799–2804

Shah (2014) reviews the impact of Management Information Systems (MIS) on school administration, emphasizing their role in enhancing efficiency and decision-making through data-driven processes (Shah, 2014, p. 2799). Shah traces the evolution of MIS from its initial use in the late 1970s for basic clerical tasks, such as storing student and personnel data, to its contemporary role in supporting strategic decision-making and operational efficiency. The study emphasizes that MIS enhances administrative functions like attendance monitoring, financial management, and resource allocation by providing timely and accurate data to school managers (Shah, 2014, p. 2800).

Key findings from the literature indicate that MIS improves decision-making processes by offering analytical tools and real-time updates, reduces administrative workload, and increases accountability to stakeholders (Shah, 2014, pp. 2801-2802).

Shah's review draws on various studies, such as Visscher (1996b), who highlights MIS's role in informed planning and evaluation, and Gurr (2000), who notes its transformative effect on leadership and communication.

The study also identifies positive outcomes, such as time savings and improved report quality, alongside challenges like inadequate training, lack of user-friendly software, and limited data analysis

skills among administrators (Shah, 2014, p. 2802). A notable contribution is its global perspective, synthesizing research from developed and developing countries, including Malaysia and Turkey, to underscore contextual differences in MIS adoption.

However, the study has limitations. It primarily focuses on data entry and collation, reflecting a gap in the literature at the time (Shah, 2014, p. 2801). Additionally, while it calls for further research into site-specific needs and system customization, it offers little practical guidance on implementation. This work remains foundational for understanding MIS's broad impact but highlights the need for more detailed, practice-oriented studies.

- **Relevance to FET SPACE:** This study served as a foundational piece for understanding the strategic importance and broad impact of a comprehensive School Management System. It informed our approach to recognizing how an SMS extends beyond simple data storage to support administrative efficiency, decision-making, and accountability across various modules, including those related to student, staff, and timetable management, which are core to FET SPACE. Shah's identification of challenges like lack of user-friendly software and training also highlighted the importance of intuitive UI/UX design and user training in our project.

2.3.2 Gehlawat, M. (2014). School Management Information System: An Effective Tool for Augmenting the School Practices. *New Frontiers in Education*, 47(2), 57–64

Manju Gehlawat's article, published in *New Frontiers in Education* (June 2014), explores School Management Information Systems (SMIS) as practical tools for improving school operations. Gehlawat defines SMIS as systems designed to align with a school's structure, management tasks, and instructional needs, emphasizing their role in simplifying administrative processes and enhancing efficiency (Gehlawat, 2014, p. 58). The study details specific SMIS modules—such as Student Management System (SMS), Attendance Management System (AMS), Examination Management System (EMS), Staff Management System, Timetable Management System (TMS), Financial Management System (FMS), and Facilities and Assets Management System (FAMS)—and their contributions to managing student data, attendance, assessments, staff, schedules, finances, and infrastructure (Gehlawat, 2014, pp. 60-63).

Key findings include SMIS's ability to reduce paper-based work, automate repetitive tasks, and improve communication among staff and parents. For instance, the EMS module supports assessment analysis and target setting, while the FMS aids in budget planning and financial transparency

(Gehlawat, 2014, pp. 61-62). Gehlawat supports her arguments with references to prior works, such as Telem (1999), who highlights SMIS's impact on departmental roles, and Anderson and Dexter (2005), who stress the importance of technology leadership. The study asserts that SMIS not only streamlines operations but also supports strategic goals by providing a centralized platform for data management.

A significant contribution of Gehlawat's work is its practical focus, offering a detailed breakdown of SMIS components and their applications, which makes it valuable for educators and administrators seeking actionable insights. However, its limitations include a lack of empirical data or case studies to substantiate claims, relying instead on theoretical assertions and secondary sources. Additionally, it does not deeply address implementation challenges like cost, training, or user resistance, which are critical for real-world adoption. This work complements Shah's broader review by zooming in on SMIS as a specific tool, though it would benefit from more evidence-based analysis.

- **Relevance to FET SPACE:** Gehlawat's work provided a detailed breakdown of specific SMIS components and their practical applications. This was particularly valuable in defining the scope and modules for FET SPACE, such as Course Management, Assignment Management, General Announcements, and the future consideration of Timetable Generation and Attendance Tracking. Her emphasis on simplifying administrative processes and improving communication reinforced our project's goals for intuitive interface and real-time updates.

2.3.3 JayaSudha, A. R., & Arunachalam, C. P. (2023). School Management System. *Journal of Advance Research in Mobile Computing*, 5(1), 35–39

The study by Dr. A R JayaSudha and Arunachalam C. P., published in the *Journal of Advance Research in Mobile Computing* (2023), presents the development of an Online School Management System using PHP CodeIgniter. This web-based application focuses on maintaining accurate records of student activities, including data on instructors, courses, students, fee payments, disciplines, and lesson details, while also managing punctuality and guardian interactions (JayaSudha & Arunachalam, 2023, p. 35). The system features distinct panels for administrators, students, teachers, and parents, with admins overseeing critical aspects like attendance, grades, curriculum, and timetables via a centralized database (JayaSudha & Arunachalam, 2023, p. 36). Built with HTML, JavaScript, PHP, and MySQL, and hosted on a local XAMPP server, the system aims to automate manual processes, reduce paper-based risks, and save time.

Key findings highlight the system's ability to eliminate manual data maintenance challenges, offering a faster, fully automated alternative to traditional methods. It stores all school-related data in a relational database, enhancing accessibility and reducing errors associated with physical records (JayaSudha & Arunachalam, 2023, p. 37). The study emphasizes practical advantages, such as ease of use and time efficiency, supported by a modest hardware setup (8GB RAM, Intel Core i3, 500GB hard disk). The authors position their work within the broader context of educational technology, referencing Brumbulli et al. (2011) for the system's operational support and general ERP principles for improving education (JayaSudha & Arunachalam, 2023, p. 35).

This work's primary contribution lies in its technical specificity, providing a clear methodology and implementation details that could guide similar projects. Its focus on stakeholder-specific panels enhances its practical relevance. However, limitations include its narrow scope—lacking empirical testing beyond basic data trials—and an absence of discussion on scalability, security, or user training needs (JayaSudha & Arunachalam, 2023, p. 39). Additionally, the study does not address potential challenges like system maintenance or integration with existing infrastructure, limiting its depth. While it advances the conversation on automation, it remains a proof-of-concept rather than a comprehensively evaluated solution.

- **Relevance to FET SPACE:** JayaSudha and Arunachalam's study was highly relevant for its practical implementation insights and architecture choices. Their use of a web-based application with distinct role-based access panels (Admin, Student, Teacher) directly influenced our decision to design FET SPACE with dedicated functionalities and interfaces for our three primary user groups. The focus on eliminating manual data maintenance and leveraging a relational database reinforced our system's core objective of automating Student Information Management and Course Management through a centralized platform.

2.3.4 Adesina, A. O., Ajagbe, S. A., Odule, T. J., & Agbele, K. K. (2022). Development of an Improved School Information Management System. *FUW Trends in Science & Technology Journal*, 7(1), 120–134

Ademola O. Adesina et al.'s article, published in *FUW Trends in Science & Technology Journal* (April 2022), details the design and implementation of an improved School Information Management System (SIMS) for post-primary schools in Nigeria. Motivated by issues like truancy and declining academic performance, the web-based system uses HTML, CSS, PHP, JavaScript, and MySQL to computerize student attendance and results, granting parents and caregivers access via a web platform

(Adesina et al., 2022, p. 120). The system was designed using a three-tier client-server architecture and implemented via explorative research and object-oriented analysis. Use case diagrams defined user roles (admin, teachers, parents, students) and functionalities (Adesina et al., 2022, pp. 123–125). Implemented on a local XAMPP server, it features role-based access for administrators, teachers, students, and parents, aiming to replace inefficient manual systems and enhance information dissemination (Adesina et al., 2022, pp. 123-124).

The study identifies key findings: the system reduces paperwork, boosts staff efficiency, and curbs truancy by enabling real-time monitoring of attendance and grades (Adesina et al., 2022, p. 130). It addresses Nigeria-specific challenges, such as manual record-keeping in public schools, supported by data from the Oyo State Ministry of Education (Adesina et al., 2022, p. 122). The authors draw on prior works, like Kornelis and Ock (2014) for system design principles and Manju (2014) for SMIS's role in school practices, while introducing a novel Student Evaluation of Teaching (SET) component to improve teacher accountability (Adesina et al., 2022, p. 124). Detailed use case diagrams and system architecture further enhance its technical rigor.

A major contribution is its focus on stakeholder engagement, particularly parental involvement, and its context-specific design for Nigerian schools. The SET feature adds an innovative dimension to traditional SMIS frameworks. However, limitations include its reliance on a local server, restricting scalability, and a lack of discussion on cost, internet dependency, or long-term maintenance (Adesina et al., 2022, p. 130). The study also lacks robust empirical validation beyond initial implementation, weakening its generalizability. Nonetheless, it offers a practical, localized solution that bridges gaps in previous works by emphasizing truancy and parental oversight.

- **Relevance to FET SPACE:** This work was significant for its focus on a context-specific design (Nigerian schools) and its emphasis on stakeholder engagement, particularly parental involvement. The three-tier client-server architecture resonated with our Global Architecture design. Importantly, the inclusion of a "Student Evaluation of Teaching (SET)" component informed our decision to propose Lecturer Rating as a future work, highlighting a valuable feature for continuous improvement not always found in basic SMS. Their use of use case diagrams also reinforced our system modeling approach.

2.3.5 Orhani, S., Saramati, E., & Drini, L. (2023). School Administration Through the School's Electronic Management System. *European Journal of Educational Management*, 6(1), 59–67

Senad Orhani, Emir Saramati, and Lulzim Drini's study, published in the *European Journal of Educational Management* (2023), investigates the implementation and perception of an Electronic School Management System (ESMS) at "Heronjtë e Lumës," a primary and lower secondary school in Vermica, Kosovo. The authors explore how this web-based system, featuring an online database, automates administrative tasks such as lesson planning, attendance tracking, and reporting, while facilitating communication among teachers, parents, and administrators (Orhani et al., 2023, p. 61). Using a mixed-method survey approach, the study gathers quantitative and qualitative data from 25 teachers via a questionnaire, analyzed with SPSS and narrative interpretation, to assess the system's impact on school administration (Orhani et al., 2023, pp. 62-63).

Key findings reveal overwhelmingly positive teacher perceptions: 92% found the system useful, 96% noted improved management processes, and 88% supported its continued use, citing benefits like enhanced planning, reduced paperwork, and faster information access (Orhani et al., 2023, pp. 62-63). Statistical analyses (ANOVA and Pearson correlation) confirmed no significant influence of gender, age, experience, or tech knowledge on opinions, with a strong positive correlation ($r = .716$, $p < .01$) between teacher views and system integration (Orhani et al., 2023, pp. 64-65). Qualitative feedback highlighted its alignment with Kosovo's Education Strategy 2022-2026 and its role in positioning the school as a regional model, though some teachers noted initial adaptation difficulties and a need for training (Orhani et al., 2023, p. 64).

The study's contribution lies in its empirical evaluation of ESMS in a specific context, blending statistical rigor with practical insights, and reinforcing literature on digital administration's efficiency (e.g., Pekisheva, 2021; Setyawan, 2021). However, its limitations include a small sample size (25 teachers from one school), limiting generalizability, and a lack of detailed technical discussion on system design or scalability (Orhani et al., 2023, p. 66). This work provides a valuable case study but calls for broader research to validate its findings across diverse settings.

- **Relevance to FET SPACE:** This study provided crucial insights into user perception and the benefits of electronic systems from the perspective of teaching staff. The overwhelmingly positive teacher feedback validated our pursuit of an intuitive and efficient system, emphasizing the importance of responsive design and real-time updates to gain user

acceptance. It also subtly underscored the need for adequate user training during system rollout, a non-functional consideration for successful adoption of FET SPACE.

2.3.6 Grepon, B., Baran, N., Gumonan, K., Martinez, A., & Laca, M. (in press). Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines

Grepon et al. (2021) examine the design and implementation of a School Management Information System (SMIS) developed for Northern Bukidnon State College in the Philippines, a context marked by inefficient manual processes and fragmented data systems. The aim was to create a centralized, network-based platform capable of streamlining core administrative functions, including admission, enrollment, assessment, grading, and reporting.

The system was developed using an Agile methodology, emphasizing iterative development and responsiveness to institutional needs. Requirements were gathered through interviews with staff familiar with office transactions and through an analysis of institutional forms, which helped shape the user interface and data workflows. The system architecture employed a client-server model under a Star topology network configuration, with design considerations addressing database structure, user roles, and interface layout.

A hybrid deployment strategy was adopted, combining a Java-based platform for handling transactional operations and a web-based platform for real-time information queries such as grade summaries and student data. Quality assurance followed ISO 25010:2011 standards, and system evaluation involved nine purposively selected users (including registrar staff and instructors), who assessed the platform across dimensions of functionality, usability, and reliability using a 5-point Likert scale.

The local implementation was hosted on a server within the administrative building, linking departmental offices through basic networking infrastructure. Provisions for remote upgrades, data backup, and restoration were included to ensure system continuity and data integrity.

The system supported multiple user roles with distinct access privileges. Super Admin users were responsible for core administrative configurations and reporting functions; registrar and program staff managed enrollment-related activities and student records; instructors uploaded and reviewed student grades through either manual input or Excel file integration. Key system features included modules for enrollment, class scheduling, grade viewing, dashboard summaries, and faculty loads—accessible

through both Java and web-based interfaces. The dashboard allowed real-time monitoring of enrollment data, while the reporting functionalities facilitated regulatory compliance and institutional decision-making.

In terms of performance, the system achieved a mean rating of 4.15 for both functionality and usability, suggesting a high degree of task completeness, accuracy, and ease of use. Reliability scored slightly lower at 3.83 but remained within a “very good” range, resulting in an overall average of 4.04. These findings indicate a positive user experience and effective system functionality in supporting school management operations.

Although the study does not formally present a conclusion, it highlights improvements in enrollment processing and data accessibility. The centralized system structure helped reduce administrative burden, and the hybrid deployment model proved adaptable to complex academic workflows. The authors also point to future enhancements such as mobile-based attendance tracking, on-campus kiosks for student access, and the development of a more comprehensive online student information portal.

This study offers a practical framework for similar institutions aiming to digitize administrative processes through modular, network-based systems. It aligns with previous work by Balcita and Palaoag (2020), who also emphasized the role of integrated platforms in improving data handling, and by Durnali (2013), who examined the impacts of digital systems on institutional efficiency. However, limitations are acknowledged, including the local-only hosting model, which restricts online accessibility, and a small evaluation sample, which may limit generalizability. Additionally, technical specifics such as the Java framework or database configuration are underreported, and the lack of cloud integration may pose challenges for scalability.

Overall, Grepon et al.'s contribution underscores the relevance of hybrid SMIS solutions in resource-constrained educational contexts and suggests potential directions for future development and evaluation.

- **Relevance to FET SPACE:** This study offered valuable insights into system design and evaluation methodologies. Their adoption of the Agile methodology for iterative development aligns with best practices for software development, influencing our project's flexible approach. More importantly, their evaluation against ISO 25010:2011 standards for functionality, usability, and reliability provided a benchmark for the quality attributes we aimed for in FET SPACE, particularly our non-functional requirements such as Performance, Reliability, Scalability, and Intuitive Interface. The use of hybrid

system architecture also provided a reference point for future architectural considerations for FET SPACE.

2.3.7 Caratiquit, K. D. (2021). Web-based School Information and Publication System: A Developmental Study

Kevin D. Caratiquit's study, published in the *Global Education & Social Sciences Journal* (2021), details the development and evaluation of a Web-based School Information and Publication System for Lal-lo National High School in Cagayan, Philippines. The system aimed to promote the school online, deliver timely information to stakeholders (staff, learners, parents, and community), share activity updates, and provide downloadable instructional resources (Caratiquit, 2021, p. 45). Developed using the iterative waterfall methodology, the system was assessed by 126 teaching and non-teaching staff, 350 learners, 70 parents, and 12 IT specialists against ISO 25010:2011 standards, including functional suitability, reliability, performance efficiency, usability, security, compatibility, maintainability, and portability (Caratiquit, 2021, pp. 47-48).

Findings indicate an overwhelmingly positive reception, with an overall weighted mean of 4.29 to 4.31 across all criteria, rated "Excellent" on a 5-point scale (Caratiquit, 2021, pp. 48-54). Specific strengths include effective functionality (e.g., mean 4.39 for helpful information), reliable and clear content delivery (mean 4.29), efficient multi-user performance (mean 4.26), and robust security features like access levels and password protection (mean 4.31) (Caratiquit, 2021, pp. 48-52). The system's compatibility with various devices (mean 4.29) and ease of maintenance (mean 4.31) further enhances its utility, supporting real-time updates and accessibility (Caratiquit, 2021, pp. 52-54). Caratiquit aligns these outcomes with prior work, such as Doctor (2017), emphasizing cost, time, and effort savings, and Koomsin (2018), highlighting improved parental engagement (Caratiquit, 2021, pp. 46-47).

The system was deemed valuable and effective, enabling concurrent multi-user functionality, robust security with database logs and access controls, and compatibility with diverse standards and devices. Its comprehensive, user-friendly design, supported by detailed guides, ensured accessibility via local networks and across workstations (Caratiquit, 2021, p. 54). Caratiquit aligns these outcomes with prior work, such as Doctor (2017), which emphasized cost, time, and effort savings through web-based systems, and Koomsin (2018), which highlighted improved parental engagement through accessible platforms (Caratiquit, 2021, pp. 46-47).

This study contributes a comprehensive evaluation of a web-based system tailored to a high school context, emphasizing stakeholder connectivity and resource accessibility. Its use of ISO 25010 provides a rigorous quality benchmark, making it a practical model for similar institutions. However, limitations include its focus on a single school, potentially limiting generalizability, and a lack of discussion on development costs or long-term sustainability challenges (Caratiquit, 2021, p. 54).

Caratiquit recommends several practical steps to enhance the system's adoption and sustainability: (1) Assigning a permanent ICT teacher to guide staff and oversee maintenance, ensuring consistent operation; (2) Encouraging the school to maintain and sustain the system to improve its information and publication processes; (3) Upgrading the web hosting plan to enhance performance and features; and (4) Establishing a quality assurance team to ensure the accuracy and quality of published information (Caratiquit, 2021, p. 54).

This work underscores the value of web-based platforms in education, particularly for stakeholder engagement and resource sharing, but suggests further research into broader applications, such as multi-school deployments, and technical enhancements, such as real-time communication or mobile app integration, to meet evolving educational needs.

- **Relevance to FET SPACE:** Caratiquit's work provided a strong example of a comprehensive web-based solution for information dissemination and stakeholder engagement. Its evaluation against the full spectrum of ISO 25010:2011 quality characteristics (including functional suitability, reliability, performance efficiency, usability, security, compatibility, maintainability, and portability) served as a guide for defining and assessing the non-functional requirements of FET SPACE. The emphasis on real-time updates, secure access levels, compatibility with various devices, and ease of maintenance directly informed our design priorities for the FET SPACE platform. It also highlights the importance of providing downloadable resources for offline access features.

2.4 Partial Conclusion

In conclusion, the review of existing literature underscores the indispensable role of School Management Systems (SMS) in modernizing educational administration. These systems, as evidenced by studies like Orhani et al. (2023) and Caratiquit (2021), enhance efficiency, automate administrative tasks, and foster collaboration among stakeholders, aligning with broader educational goals of transparency and performance improvement (Shah, 2014; UNESCO, 2022). The evolution of SMS technologies—from basic data management tools to sophisticated platforms supporting real-

time analytics and stakeholder engagement—demonstrates their potential to transform school operations (Grepon et al., in press; Gehlawat, 2014). However, challenges persist, including scalability limitations, infrastructure costs, and the need for user training, as noted in works like Adesina et al. (2022) and JayaSudha & Arunachalam (2023).

The analysis of related works reveals a range of benefits and trade-offs across different SMS implementations. Locally hosted systems, such as those in Grepon et al. (in press), offer robust control but lack online accessibility, while web-based solutions like Caratiquit (2021) excel in stakeholder connectivity yet demand reliable internet infrastructure. Automation and centralization, as seen in JayaSudha & Arunachalam (2023), reduce manual workloads, yet their effectiveness hinges on system quality and user adoption (Orhani et al., 2023). Understanding these dynamics is essential for designing SMSs that balance functionality with practicality, ensuring they meet diverse institutional needs. Despite these promising advancements, there remains a paucity of research on integrating emerging technologies into SMS frameworks, presenting a significant opportunity for this study.

Addressing issues like user resistance, system reliability, and scalability will be critical for enhancing SMS effectiveness. Continuous technological advancements, integration with educational workflows, and improved usability—echoed in online insights from EdTech Magazine (2023)—are necessary to keep pace with evolving administrative demands. This research aims to bridge these gaps by proposing an SMS that leverages the theoretical foundations and practical insights from this review, contributing to the ongoing development of educational management tools. The next chapter will outline the methodology for designing and validating this proposed system, building on the strengths and addressing the limitations identified herein.

CHAPTER 3: ANALYSIS AND DESIGN

3.1. Introduction

This chapter presents the methodology and design approach adopted in developing the SMS. It outlines the major phases of the methodology, describes the requirements gathering and analysis process in detail, and explains the design techniques and tools used in modeling the system. It also covers the system architecture and highlights the design decisions made throughout the project.

3.2. Methodology

The project follows the Agile methodology, which promotes iterative development, flexibility, and continuous collaboration with stakeholders. The methodology is organized into the following phases:

- **Requirements Gathering:** Collecting input from stakeholders to define system needs.
- **Requirements Analysis:** Prioritizing and classifying requirements into functional and non-functional categories.
- **Design:** Developing system architecture, modeling, and user interface (UI) design.
- **Development and Iterative Implementation:** Building the system feature by feature.
- **Integration and Testing:** Combining components and validating functionality and usability.

3.3. Requirements Gathering and Analysis

3.3.1. Requirements Gathering

Requirements were gathered from students and teachers to ensure the system meets the needs of both groups.

- **Students:** a survey was sent out to students to collect their preferences and expectations. The survey questions are available in Appendix A.
- **Teachers:** in person interviews were conducted with faculty members. The process began by presenting the most frequently requested features from students to align expectations. This was followed by open-ended questions such as:
 - “What would you like to see on this platform?”
 - “What student details do you consider important?”
 - “What device do you primarily use to access such platforms?”

Summary of Teacher Interview Feedback:

Table 1: Summary of teacher requirements gotten after interview

Stakeholder	Key Needs	Access Preference
Dr. Tsague (Lecturer)	General announcements, auto-generated questions from uploaded materials, student profiles	Laptop
Dr. Nguti (Lecturer and Admin)	Past questions bank, add students via link/manual, general and course announcements	Laptop (primary) Phone (secondary)
Dr. Valerie (Lecturer and Admin)	Timetable generation, attendance tracking, graduation alerts, admin dashboard	Laptop & Phone
Dr. Fendji (Lecturer)	Attendance tracking (minimum 70%), course-specific announcements, general and departmental timetables	Phone
Dr. Sop (Lecturer and Admin)	Bulk student addition (CSV/Excel), assign staff to add students, upload past tutorials/questions	Laptop
Dean	Strong emphasis on timely announcements, events, and timetables	Not specified

Table 1 provides a summary of the teachers' requirements identified during the interview.

The goal was to capture both operational requirements and platform usage preferences. The list of requirements from both groups is as follows:

- **Students:**
 - General announcements
 - Course-specific announcements
 - Course management (e.g., access to materials, revision questions, schedules)
 - Assignment submission
 - Offline Access
 - Group chats
 - GPA calculator
 - Transcript Application
 - About FET
 - Location tracking
 - Simple and intuitive user interface
- **Teachers:**
 - General announcements (e.g., timetables, events)
 - Course-specific announcements (e.g., CA marks, course schedules)
 - Course Management (e.g. material upload)
 - Assignment management
 - Student profile access: name, matricule, department, level, and picture
 - Attendance tracking
 - Timetable generation and access (general and department-specific)
 - Timetable visibility (global and course-specific)
 - Past questions and tutorial uploads
 - Question generation based on course material
 - Student addition (manual or via CSV/Excel upload)
 - Responsive interface for laptop and phone access
 - Centralized system to replace fragmented tools like WhatsApp

3.3.2. Requirements Analysis

The gathered requirements were analyzed to prioritize and classify them based on relevance, demand, and feasibility within the project's constraints.

Which of these features do you absolutely want in FET Space? (Select all that apply)

85 responses

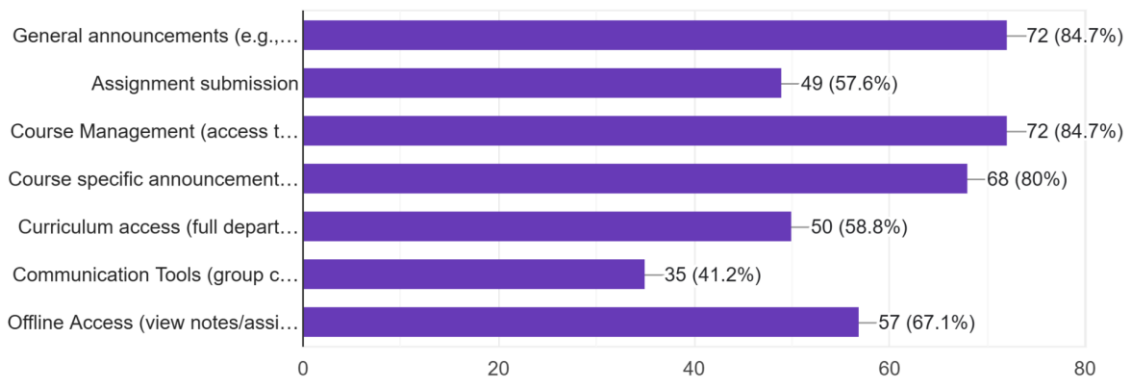


Figure 1: Student preferences for platform features considered essential for future development.

Fig 1 presents a bar chart illustrating the survey results, where students selected their most desired features for the system. Combined with recurring and frequently requested needs expressed by lecturers and administrators, these insights informed the selection of the following requirements for inclusion in the School Management System:

- General announcements
- Course management
- Assignment management
- Offline access
- Simple and intuitive user interface
- Responsive interface for laptop and phone access
- Student profile access (name, matricule, department, level, picture)
- Timetable visibility (global and course-specific)
- Past questions and tutorial uploads
- Student addition (manual or via CSV/Excel upload)
- About FET

The following features were left out at this stage

- Location tracking
- GPA calculator
- Group chats
- Timetable generation and access (general and department-specific)

- Attendance tracking
- Question generation based on course material
- Transcript application

Based on the chart in Fig 2 from the survey and interviews, the mobile app currently includes some teacher functionalities but is primarily designed for students, while the web platform mainly supports teacher features. Both platforms will be enhanced in future updates to fully support all user groups.

Do you prefer accessing FET Space through a mobile app or a web platform?
85 responses

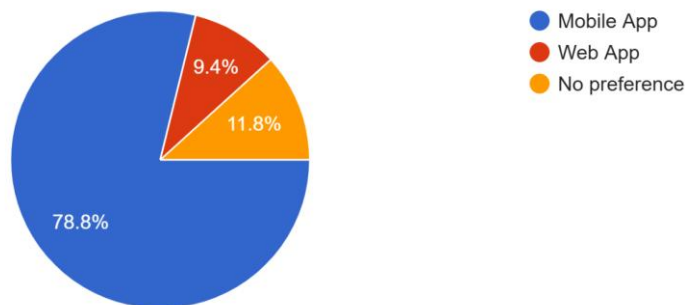


Figure 2: Pie chart showing the students preferred platform access methods

Functional and Non-Functional Requirements

The requirements were then classified into functional and non-functional requirements as seen below:

Functional Requirements:

- General and course-specific announcements with real-time delivery
- Course management (material access, updates, offline support)
- Assignment management (creation, submission, tracking)
- Student information management (view by admins/teachers)
- Curriculum access (online and offline)
- Timetable generation and access
- Past question and tutorial uploads
- Question generation from course material
- Student addition (manual or CSV/Excel)

Non-Functional Requirements:

- Simple, minimalistic, and intuitive interface
- Responsive design for web (teachers) and mobile (students)
- Real-time updates
- Performance
- Reliability
- Scalability
- Offline access for key features
- Smooth animations and transitions
- Secure and centralized platform

Prioritized Requirements

Based on the student survey results and the frequency of requirements gathered from students, teachers, and administrative staff (i.e., how often specific needs were mentioned during the requirements gathering process), the system requirements were prioritized as follows:

- **Priority 1 Features:**
 - General Announcements
 - Course-Specific Announcements
 - Course Management
 - Design Principles (simple, minimalistic interface; smooth animations)
- **Priority 2 Features:**
 - Assignment Management (create, view, submit assignments)
 - Curriculum Access (school curriculum, offline access)
 - Offline Access (announcements, curriculum, notes, past questions)
- **Priority 3 Features:**
 - Assignment Reminders
 - About FET (information about the Faculty of Engineering and Technology)
- **Priority 4 Features:**
 - Class Reminders (notifications for class schedules)
 - Multilingual Support (multi-language interface)

3.4 Design

The design phase focused on modeling the system and creating an intuitive user interface.

3.4.1 System Modeling

System modeling was conducted using several key diagrams to provide a clear and organized representation of the system from different perspectives. These diagrams include the Use Case Diagram, which outlines user interactions and system functionalities; the Class Diagram, detailing the system's static structure and relationships; the Sequence Diagram, showing the flow of interactions over time; and the Entity Relationship Diagram (ERD), which defines the database structure. Together, these models ensure a comprehensive understanding of both the system's behavior and data organization.

3.4.1.1. Use Case Diagram

The use case diagram in Fig 3 maps the functional requirements of the School Management System, highlighting how different user types (Students, Teachers, Admins) interact with various system features. The diagram includes:

- **Actors:** Students, Teachers, and Admins.
- **Use Cases:** Specific actions or functionalities each actor can perform.
- **Relationships:** Lines connecting actors to use cases, indicating permissible interactions.

Actors and Their Roles

- **Student:**
 - Represents learners who engage with the system to access course-related information and submit work.
 - Primary focus is on self-managed course participation and assignment submission.
- **Teacher (lecturers):**
 - Represents educators responsible for managing courses, content, and student interactions.
 - Acts as an intermediary between students and admins, with enhanced control over course materials and assessments.
- **Admin:**

- Represents system administrators or higher-level staff who oversee user management, announcements, and overall system operations.
- Holds the highest level of access, including managing other users.

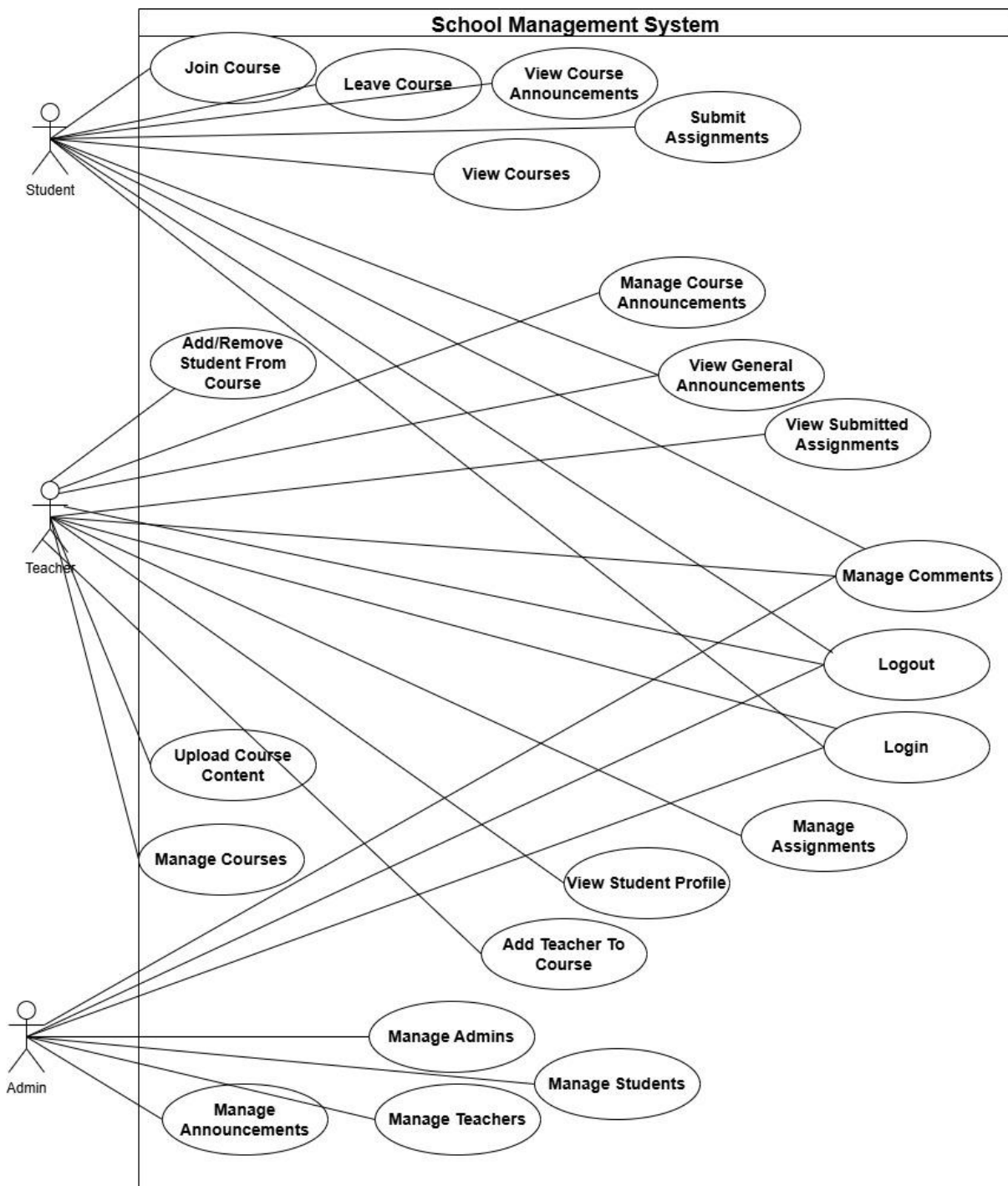


Figure 3: Use case diagram for the school management system

[update the diagram to ensure all the functionalities should have login as an include]

Use Cases and Descriptions

Below is a detailed breakdown of each use case, grouped by actor, with explanations of their purpose and relevance.

Note: “Manage” in this context includes the ability to add, edit, delete, and view records.

Student Use Cases

- **Join Course:** Enroll in a course to access its materials and announcements.
- **Leave Course:** Withdraw from a course and lose access to its resources.
- **View Course Announcements:** See updates related to enrolled courses.
- **Submit Assignments:** Upload completed assignments for grading.
- **View Courses:** Display all enrolled courses, including schedules and materials.
- **View General Announcements:** Access system-wide announcements from admins.
- **Manage Comments:** Add feedback on assignments or announcements.
- **Login/Logout:** Securely access or exit the system.

Teacher Use Cases

- **Upload Course Content:** Add materials such as notes, past questions, and tutorials.
- **Manage Courses:** Add, edit, or remove courses they created or were added to and update course information.
- **Manage Course Announcements:** Post or update announcements related to a course.
- **View General Announcements:** Access system-wide announcements from admins.
- **View Submitted Assignments:** Review and grade student submissions.
- **Manage Comments:** Add feedback on assignments or announcements.
- **View Student Profiles:** Access student details like name and matricule.
- **Add/Remove Student from Course:** Add or remove students from course so they can access or not access materials, announcements etc.
- **Manage Assignments:** Add, edit, delete, or update course assignments across all their courses.
- **Add Teacher to Course:** Assign other teachers to help manage a course.
- **Login / Logout:** Securely access or exit the system.

Admin Use Cases

- **Manage Announcements:** Create, edit, delete, and update system-wide announcements.
- **Manage Teachers:** Add, edit, or remove teacher accounts and roles.

- **Manage Students:** Add, edit, or remove student accounts and enrollments.
- **Manage Comments:** Add feedback on announcements.
- **Manage Admins:** Handle admin roles and add or remove admin accounts.
- **Login / Logout:** Securely access or exit the admin system.

3.4.1.2. Class Diagram

The Class Diagram serves as a foundational component of the system modeling phase, offering a static, structural view of the School Management System. It visually represents the system's core entities (classes), their attributes (data they hold), the operations they can perform (methods), and the various relationships that exist between them. This diagram is crucial for understanding the system's architecture, data organization, and the interactions between different components, laying the groundwork for database design and software implementation.

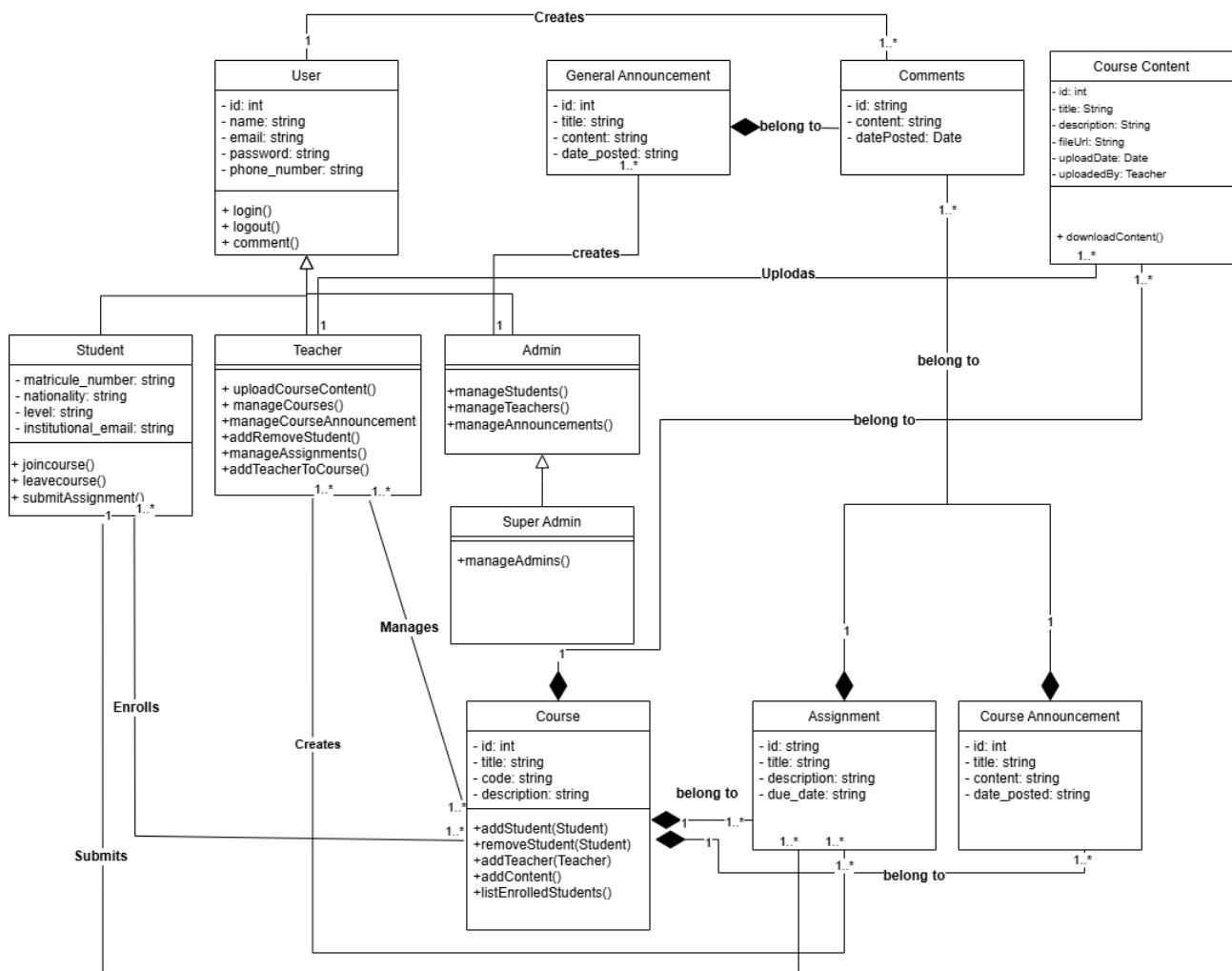


Figure 4: Class diagram for the school management system

Classes, Attributes, and Methods

From Fig 4, the SMS is composed of the following key classes.

Note: For methods prefixed with "manage," this implies the ability to perform full CRUD (Create, Read, Update, Delete) operations.

- i. **User (Abstract)**
 - **Purpose:** Base class for all system users.
 - **Attributes:** id, name, email, password, phone_number.
 - **Methods:** login(), logout(), comment().
- ii. **Student**
 - **Purpose:** Learner role, inherits from User.
 - **Attributes:** matricule_number, nationality, level, institutional_email.
 - **Methods:** joinCourse(), leaveCourse(), submitAssignment().
- iii. **Teacher**
 - **Purpose:** Educator role, inherits from User.
 - **Methods:** uploadCourseContent(), manageCourses(), manageCourseAnnouncement(), addCourseStudent(), removeCourseStudent(), addTeacherToCourse().
- iv. **Admin**
 - **Purpose:** System administrator, inherits from User.
 - **Methods:** manageStudents(), manageTeachers(), manageAnnouncements().
- v. **Super Admin**
 - **Purpose:** Top-level administrator, inherits from Admin.
 - **Methods:** manageAdmins().
- vi. **General Announcement**
 - **Purpose:** System-wide announcements.
 - **Attributes:** id, title, content, date_posted.
- vii. **Course Content**
 - **Purpose:** Educational materials (notes, past questions, tutorials) for courses.
 - **Attributes:** id, title, description, fileUrl, uploadDate, uploadedBy (Teacher).
 - **Methods:** downloadContent().
- viii. **Comments**
 - **Purpose:** Feedback or remarks on announcements/assignments.
 - **Attributes:** id, content, datePosted.
- ix. **Course**
 - **Purpose:** Represents an academic course.

- **Attributes:** id, title, code, description.
 - **Methods:** addStudent(), removeStudent(), addTeacher(), listEnrolledStudents(), addContent().
- x. **Assignment**
- **Purpose:** Academic tasks assigned within a course.
 - **Attributes:** id, title, description, due_date.
- xi. **Course Announcement**
- **Purpose:** Announcements specific to a particular course.
 - **Attributes:** id, title, content, date_posted.

Relationships and Cardinalities

The diagram depicts three primary relationship types:

- i. **Generalization (Inheritance):**
 - Student, Teacher, Admin are specializations of User.
 - Super Admin is a specialization of Admin.
- ii. **Associations (General Connections):**
 - **Teacher - Course:** Many-to-Many (1..* - 1..*). A teacher can teach one or more courses, and a course can be taught by one or more teachers.
 - **Student - Course:** Many-to-Many (1..* - 1..*). A student can enroll in one or more courses, and a course can have one or more students.
 - **Student - Assignment:** One-to-Many (1 - 1..*). A student submits one or more assignments.
 - **Teacher - Assignment:** One-to-Many (1 - 1..*). A teacher creates one or more assignments.
 - **Teacher - Course Content:** One-to-Many (1 - 1..*). A teacher uploads one or more content items.
 - **User - Comments:** One-to-Many (1 - 1..*). A user creates one or more comments.
- iii. **Composition (Strong Ownership):**
 - **Course - Course Content:** One-to-Many (1 - 1..*). Course Content cannot exist without its Course.
 - **Course - Assignment:** One-to-Many (1 - 1..*). Assignment cannot exist without its Course.
 - **Course - Course Announcement:** One-to-Many (1 - 1..*). Course Announcement cannot exist without its Course.

- **General Announcement, Course Announcements, and Assignments - Comments:** One-to-Many (1 – 1..*). Comments cannot exist without related to the announcements or assignments.

3.4.1.3. Sequence Diagram

In my SMS, sequence diagrams provide a step-by-step view of how users and system components interact to carry out important operations. These diagrams help us understand how the system functions, identify possible issues, and confirm that everything communicates as expected.

I created sequence diagrams for key user flows in the SMS. This helps break down each step in a process, showing how data moves between parts of the system and who is responsible for what. This level of detail supports both development and testing by ensuring the system behaves as intended.

The three key scenarios for which the sequence diagrams was developed were:

- User Onboarding Flow
- Course Access Management
- Assignment Handling (Create, Submit, Review)

A. User Onboarding Flow

Description: The diagram in Fig 5 illustrates the complete user onboarding process within the School Management System, starting with an admin creating a new user (either a teacher or a student) and concluding with that user's initial login.

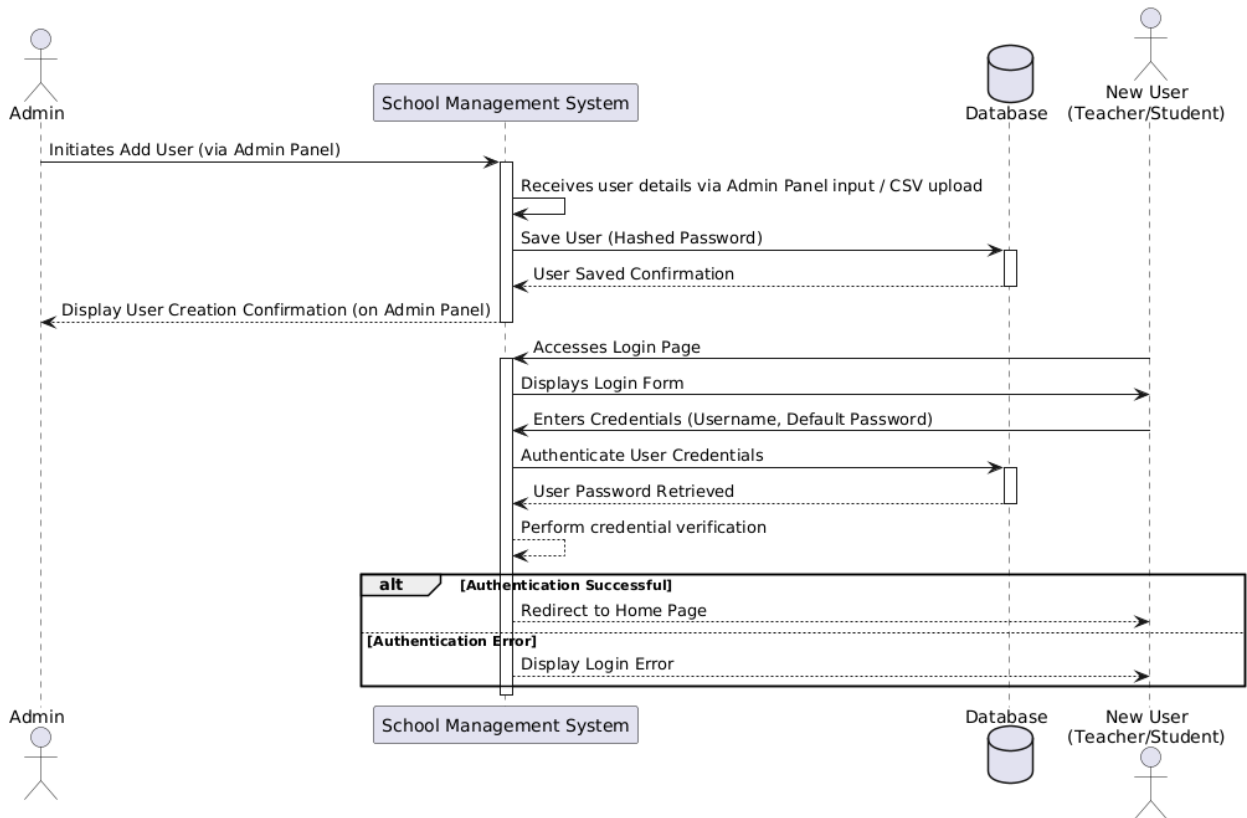


Figure 5: User Onboarding Flow

Detailed Explanation:

- **Actors/Objects:** Admin, School Management System, Database, New User (Teacher/Student).
- **Sequence Flow:**
 - The Admin adds a new user via the Admin Panel or uploads a CSV file with user details.
 - A default password is assigned (e.g., matricule for students, FETSPACETEACHERS2025 for teachers).
 - The system stores the user's data (with a hashed password) in the database.
 - The Admin receives confirmation once the user is added.
 - The new user visits the login page, enters their credentials, and submits them.
 - The system checks the credentials against the database.
 - If the credentials match, the user is logged in and redirected to their home page, but if they don't match, an error message is displayed.

B. Course Access Management

Description: This diagram in Fig 6 illustrates how a teacher can add or remove a student from a course.

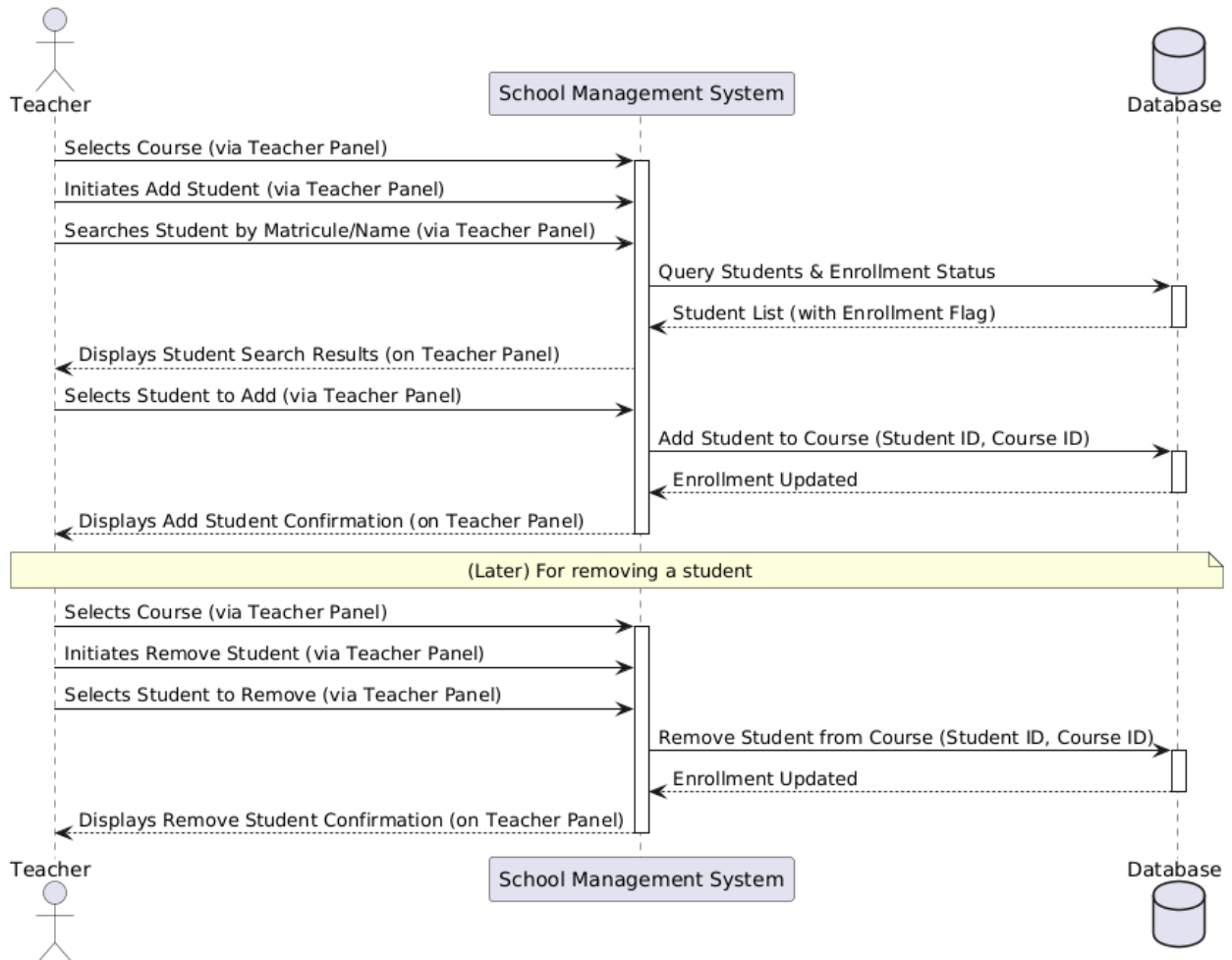


Figure 6: Course Access Management

Detailed Explanation of the Diagram:

- **Actors/Objects:** Teacher, School Management System, Database.

Sequence Flow (Adding a Student):

- The Teacher selects a course and initiates the "Add Student" action.
- They search for a student by matricule or name.
- The system queries the database to find students and check if they're already enrolled.
- Search results are displayed.

- The Teacher selects a student to add.
- The system updates the database with the new enrollment.
- A confirmation is shown once the student is added.

Sequence Flow (Removing a Student):

- The Teacher selects a course and views enrolled students.
- They choose a student to remove.
- The system updates the database to reflect the removal.
- A success message confirms the update.

C. Assignment Handling (Create, Submit, Review)

Description:

This diagram in Fig 7 covers the full assignment workflow—from a teacher creating an assignment, to a student submitting it, and the teacher reviewing those submissions.

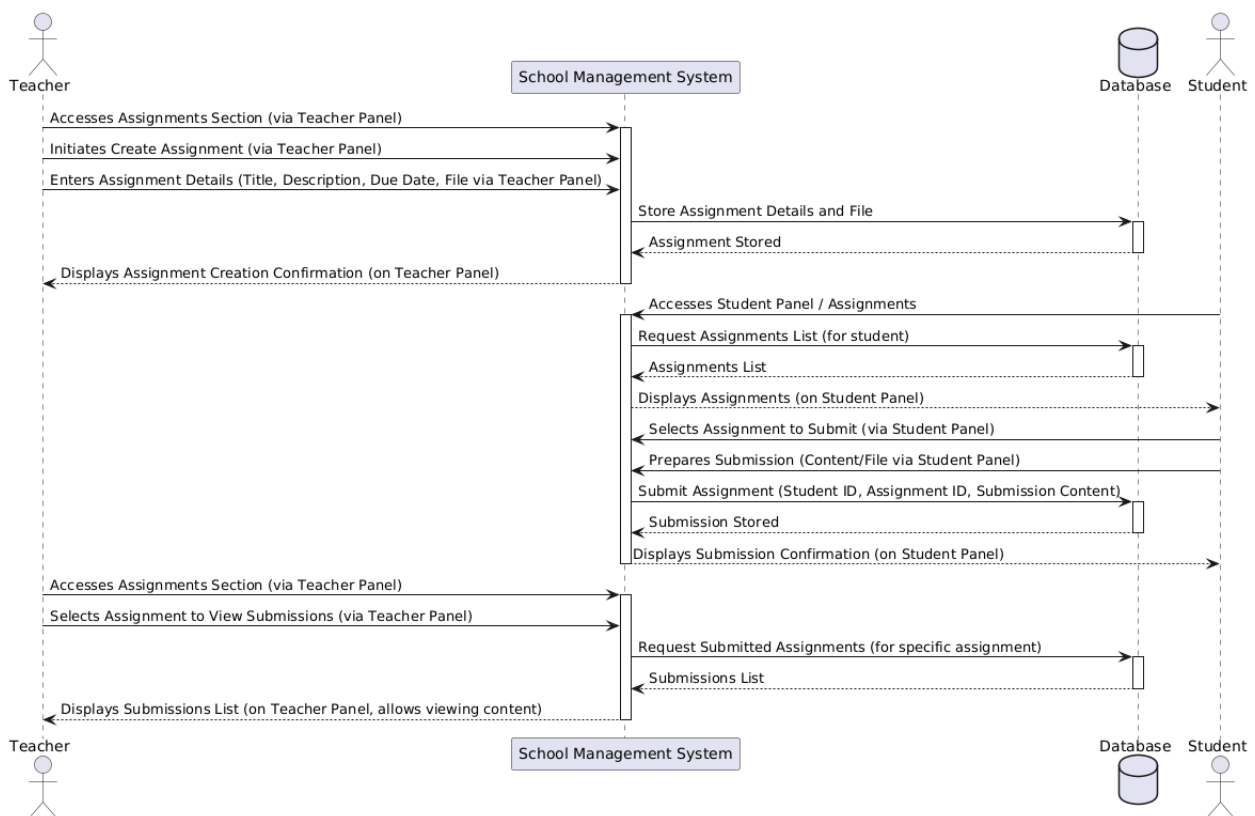


Figure 7: Assignment Handling

Detailed Explanation of the Diagram:

- **Actors/Objects:** Teacher, School Management System, Database, Student.

Sequence Flows

Assignment Creation:

- The Teacher goes to the assignments section and fills in assignment details (title, description, due date, attachments).
- The system stores this information in the database.
- The Teacher receives confirmation that the assignment was created.

Student Submission:

- The Student logs in and checks their assignments.
- The system fetches the list from the database.
- The Student selects an assignment, prepares their work, and submits it.
- The system stores the submission and updates the student's assignment status.
- A confirmation is displayed to the Student.

Teacher Review:

- The Teacher accesses the assignments section via the Teacher Panel of the School Management System.
- The Teacher selects a specific assignment to view its submissions via the Teacher Panel.
- A list of submissions (including student names and statuses) is displayed.
- The Teacher can open and review each submission individually.

3.4.1.6. Entity Relationship Diagram

The ER diagram in Fig 8 illustrates the entities and their respective attributes within the SMS, providing a visual representation of the database structure. It effectively shows the relationships between these entities, serving as a blueprint for the database design. We began by defining the different entities and their relationships with one another. Once these entities and relationships were established, we created the ER diagram to visually map them. From the ER diagram we transitioned

to converting it into a relational schema, which provided the precise structure of the database, including additional tables generated to accommodate the relationships.

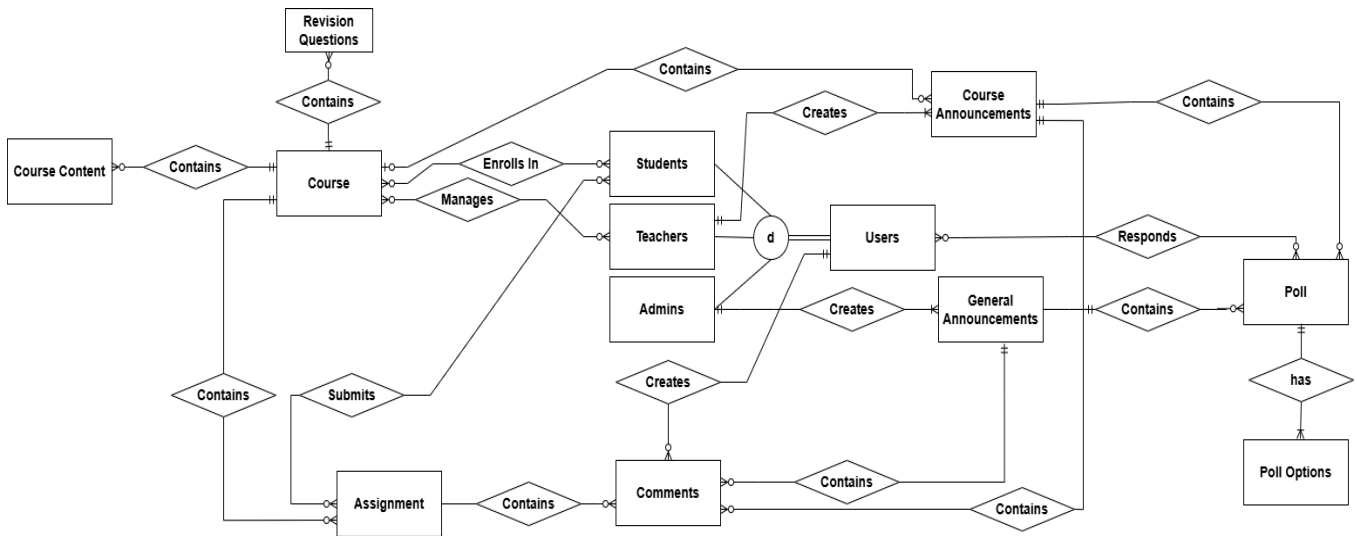


Figure 8: ER Diagram for the school management system

Initial Entities and Their Purpose

The following entities were identified as the foundational components of the system, each serving a specific role:

- **Users:** Stores all types of users (students, teachers, admins) with common information like name, email, and role.
- **Students:** Stores student-specific details such as level, matricule number, and nationality.
- **Teachers:** Stores information related to users with a teaching role.
- **Admins:** Stores data related to administrative users.
- **Courses:** Stores course-related details including title, description, and course code.
- **General Announcements:** Stores system-wide updates created by admins.
- **Course Announcements:** Stores updates posted by teachers that are specific to their courses.
- **Course Content:** Stores educational materials associated with courses.
- **Assignments:** Stores tasks created by teachers for their courses.
- **Revision Questions:** Stores revision materials available for courses.
- **Polls:** Stores surveys or voting features linked to announcements.
- **Poll Options:** Stores possible choices for each poll.
- **Comments:** Stores feedback and discussions on assignments and announcements.
- **Attachments:** Stores files attached to various entities like assignments, content, and announcements.

- **Notifications:** Stores system notifications sent to users.

Key Relationships Between Entities

The relationships between the core entities are defined as follows:

- Admins, Teachers, and Students are all users.
- A Teacher creates many Courses, but a Course belongs to only one Teacher.
- A Student enrolls in many Courses, and a Course has many Students.
- A Course contains many Course Announcements, and a Course Announcement belongs to only one Course.
- An Admin creates many General Announcements, and a General Announcement belongs to only one Admin.
- A General Announcement contains many Polls, and a Poll belongs to only one General Announcement.
- A Course Announcement contains many Polls, and a Poll belongs to only one Course Announcement.
- A Poll contains many Poll Options, and a Poll Option belongs to only one Poll.
- A User responds to many Poll Responses, and a Poll Response belongs to only one User.
- A Course contains many Course Content, and a Course Content belongs to only one Course.
- A Course contains many Assignments, and an Assignment belongs to only one Course.
- A Teacher creates many Assignments, and an Assignment belongs to only one Teacher.
- A Student submits many Assignment Submissions, and an Assignment Submission belongs to only one Student.
- An Assignment contains many Comments, and a Comment belongs to only one Assignment.
- A General Announcement contains many Comments, and a Comment belongs to only one General Announcement.
- A Course Announcement contains many Comments, and a Comment belongs to only one Course Announcement.
- A Course contains many Revision Questions, and a Revision Question belongs to only one Course.

Relational Schema Conversion

The transition from the **Entity-Relationship diagram** to the **relational schema** involved converting

entities and their relationships into database tables. This process required careful attention to **cardinality constraints**, ensuring that the schema could efficiently support real-world operations.

During this translation, several modifications were made to address **complex relationships**, particularly those involving:

- **Many-to-Many (N:M)** relationships — which necessitated the creation of **junction tables** to properly represent associations between entities (e.g., students enrolled in multiple courses and vice versa).
- **Many-to-One (N:1)** relationships — which were handled by referencing keys to support efficient joins, and simplify query operations throughout the system

Additional Tables Created Based on Relationships:

The following tables were added based on the relationship cardinality:

- **Course_Teachers:** Manages the many-to-many relationship between Teachers and Courses, storing which teachers manage which course.
- **Course_Students:** Handles the many-to-many relationship between Students and Courses, storing which students are enrolled in each course.
- **Assignment_Submissions:** Manages the many-to-many relationship between Students and Assignments, storing student submissions for assignments.
- **Poll_Responses:** Handles the many-to-many relationship between Users and Poll Options, storing user responses to polls.

These additional tables ensure the database structure accurately reflects the system's requirements, providing a robust foundation for data management and retrieval within the SMS.

3.4.2 UI Design

The UI design prioritized simplicity and usability. The process began by organizing the requirements to establish the structure of the interface, ensuring alignment with the overall system design and functional goals.

We started with wireframes as seen in Fig 9 to outline the layout and core navigation elements. These wireframes served as the foundation for the visual direction of the application.

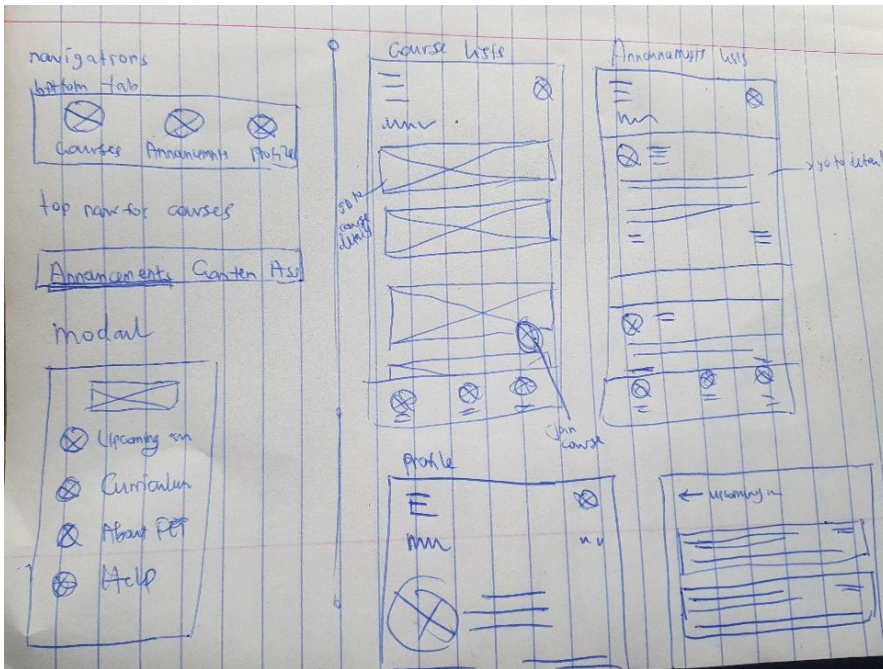


Figure 9: Wireframes of school management system

Font and Color Scheme:

We used a clean, readable font (Roboto) paired with a predominantly black-and-white interface. Blue was selectively used for subtle highlights or interactive elements, giving the system a modern and professional feel without overwhelming the user.

High-fidelity Prototypes:

Based on the wireframes, we developed the look and feel of the system in line with the defined design and requirements. The final design as seen in Fig 10 emphasized a clean, professional appearance, with high-fidelity prototypes created to reflect a responsive and intuitive interface. While we initially explored using a shade of blue, the final UI leaned more heavily on a black-and-white color scheme to achieve a refined, modern aesthetic. This choice helped reduce visual clutter and supported a more focused and user-friendly experience, incorporating feedback from wireframe reviews and usability considerations.

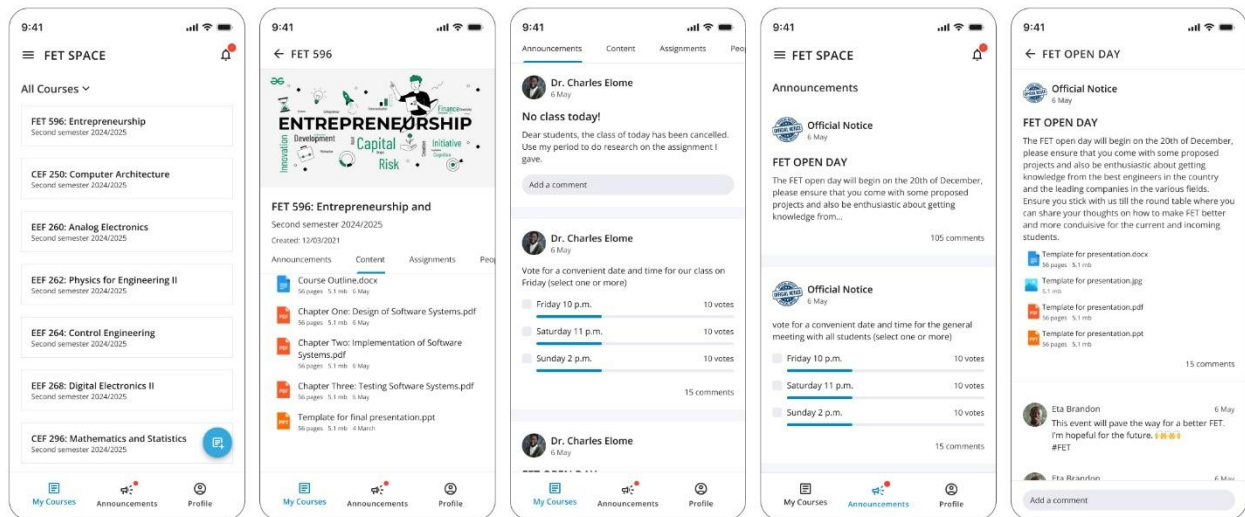


Figure 10: High fidelity prototype of school management system

3.5. Global Architecture of the Solution

The School Management System follows a multi-tier architecture designed to support role-based access, modular growth, and smooth operation across both web and mobile platforms. The architecture is composed of three primary layers: the **Client Layer**, the **Application (Server) Layer**, and the **Data Layer** as seen in Fig 11. These layers interact over secure protocols to ensure a seamless and responsive user experience.

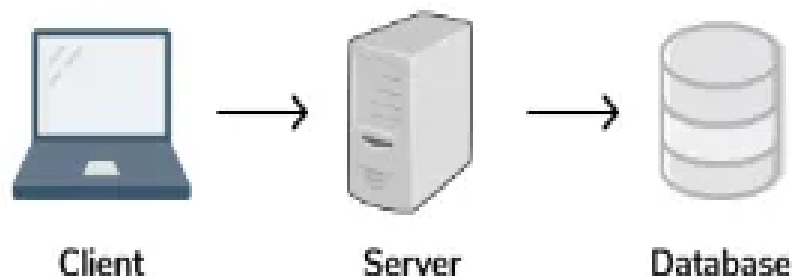


Figure 11: System Architecture Overview for school management system

3.5.1 Client Layer

The **Client Layer** serves as the interface between users and the system. It includes both mobile and web applications, tailored for different user roles.

The **mobile application**, designed for both students and teachers, provides a role-based interface that adapts its layout and functionalities depending on the authenticated user's role. It supports key offline

features like assignment access and timetable viewing, ensuring usability even without an internet connection.

The **web application** similarly supports both students and teachers, offering a responsive design for use on laptops and desktops. Like the mobile app, its interface dynamically adjusts based on the user's role, ensuring that each user type sees only the relevant tools and features.

The **admin web portal** is a separate interface exclusively for administrators. Unlike the shared student-teacher interface, this portal provides system-wide control over users, announcements, and configurations. Its separation from the main interface enhances security and maintains a clear boundary between operational tasks and content delivery.

3.5.2 Application Layer

The **Application Layer** acts as the system's core, housing the backend services responsible for executing business logic, handling user requests, and enforcing security rules.

This layer processes all incoming API calls from the client applications, managing tasks such as authentication, role-based access control, data validation, and interaction with the database. It ensures that students can view and submit assignments, teachers can manage course materials, and admins can oversee platform operations—all through a unified logic layer.

A central component of this layer is the **Role-Based Access Manager**, which determines what actions each user can perform based on their role. It ensures strict adherence to access boundaries, preventing unauthorized data access or manipulation.

Authentication is managed through a secure **authentication module**, handled using a token-based system. This ensures safe login and session handling for all users.

3.5.3 Data Layer

The **Data Layer** is responsible for securely storing and managing all system data. This includes user profiles, course content, assignments, announcements, timetables, and uploaded documents.

All data is stored in a centralized database, structured for reliability and optimized for performance. Tables or collections are designed to reflect the core entities of the system (e.g., users, courses, comments), supporting fast queries and consistent data integrity.

3.6. Partial Conclusion

This chapter gave an overview of the methodology and architectural layout of the SMS. By adopting the Agile methodology, we ensured that the development process was flexible and iterative, allowing for continuous feedback from stakeholders, and also helped us refine the system's features and design at every stage, ensuring it closely aligned with the needs and expectations of the users.

By carefully organizing and refining the requirements, we laid the foundation for creating a user-friendly interface and a robust system architecture. A strong emphasis on usability in the design ensured that the platform would be intuitive for students and educators alike. Additionally, the architectural decisions—such as adopting a three-tier design for the web and mobile interfaces—enabled scalability, and smooth integration between the frontend and backend.

With clear user requirements and a strong architectural foundation in place, we set the stage for a successful implementation. Proper planning and design also helped mitigate potential risks, ensuring the system would be both user-centric and technically feasible.

The next chapter will cover the implementation details, providing an in-depth look at how the system was built, the development process, as well as the testing strategies employed to ensure the system's functionality and reliability.

CHAPTER 4. IMPLEMENTATION AND RESULTS

4.1. Introduction

This chapter presents the results obtained after the realization of the School Management System, including screenshots and detailed explanations for each result. It outlines the deployment and execution process for the solution and evaluates its effectiveness through simulation and comparison with existing systems, demonstrating its alignment with the objectives outlined in Chapter 1 and its significant contribution to engineering and technology.

4.2. Tools and Materials Used

Client-Side Tools

Web:

- **React.js** – A JavaScript library used for building the web interface, chosen for its component-based architecture and flexibility.
- **Tailwind CSS** – A utility-first CSS framework that enabled rapid and consistent styling across the web application.
- **Shadcn UI** – Used for creating interactive UI elements like pop-ups, dialogs, and modals, enhancing the user experience with ready-made, customizable components.

Mobile:

- **React Native with Expo** – Used for developing the mobile application. Expo simplified setup and deployment, while React Native ensured cross-platform compatibility across iOS and Android devices.

Server-Side Tools:

- **Node.js with TypeScript** – Node.js served as the backend runtime, and TypeScript was used for type safety and maintainability.
- **Express.js** – A lightweight web framework for Node.js, chosen for its simplicity and flexibility in building RESTful APIs.
- **Prisma** - An ORM tool for Node.js, used to interact with SQL databases. It provides type-safe database queries, schema migrations, and a modern query API.

Testing Tools:

- **Thunder Client** – A lightweight and fast API testing tool used within VS Code to test backend endpoints during development.

Database:

- **PostgreSQL (Supabase)** – A robust and open-source relational database system selected for its reliability, scalability, and strong support for complex queries.

4.3. Description of the implementation process

The implementation process followed a structured approach—from setting up the development environments to building out the application, integrating backend services, and managing state efficiently. The goal was to ensure a seamless experience across student, teacher, and admin panels.

4.3.1 Environment Setup

Development began by setting up environments for both **web and mobile platforms**:

- The student and teacher interfaces shared a single codebase, built with React.js for web and React Native for mobile. Role-based logic was implemented to dynamically switch between student and teacher views.
- The admin interface was developed in a separate React.js codebase to accommodate its distinct administrative features and layout.

Key libraries and tools installed included:

Tailwind CSS, React Query, Axios, Shadcn UI, Prisma ORM.

Each environment was configured to promote modular and reusable component design, in alignment with React's core philosophy.

4.3.2 Frontend Development

The frontend was implemented by following the pre-designed Figma screens. This design-first approach helped ensure accurate visual alignment with the intended user experience.

Key development highlights:

- A single shared codebase was used for students and teachers, with view logic controlled by role-based conditions.
- Components were structured for maximum reusability, including shared inputs, modals, tabs, and layouts.
- Shadcn UI was used for more complex or interactive components like:
 - Modals (e.g., course joining or content creation)
 - Dialogs (e.g., delete confirmations)
 - Popovers and dropdowns (e.g., user options and context menus)

Frontend-backend integration was carried out by:

- Fetching data provided by the server (via API endpoints)
- Dynamically displaying and updating this data in the UI
- Handling submissions and mutations such as assignments or announcements, which were then passed back to the server and persisted in the database

4.3.3 Data Management with React Query

To manage data fetching and server state, **React Query** was used throughout the project. It simplified the logic for interacting with the API and offered several advantages:

- Built-in caching reduced redundant requests and improved performance
- Background fetching kept data updated without disrupting the user
- Simplified loading and error handling, reducing boilerplate in components
- Automatic retries and refetching ensured data reliability
- Optimistic updates enhanced perceived responsiveness for write operations

Typical use cases included:

- Fetch and cache course lists, user profiles, assignments, and announcements
- Submit data such as assignment uploads and comments
- Reactively update the UI based on server-side changes

4.3.4 Backend Development

The backend was built with **Node.js**, **Express.js**, and **TypeScript** for improved type safety and maintainability.

- A RESTful API architecture was followed, with endpoints covering authentication, course management, announcements, assignments, notifications, and admin actions.
- **PostgreSQL** was chosen as the primary relational database for its ability to handle structured, relational data.
- API routes were tested iteratively using **Thunder Client**, ensuring quick debugging and endpoint validation during development.

The backend served as the central logic and data hub, powering all features and interactions initiated from the frontend.

4.4. Presentation and Interpretation of Results

The application is divided into three main user panels: Student Panel, Teacher Panel, and Admin Panel. Each user type has specific features and interface views tailored to their roles.

- The Student Panel is available on both mobile (app) and web (browser), but this section focuses on the mobile view, with the web view omitted as they closely resemble the teacher web interface.
- The Teacher Panel is also accessible via both web and mobile, but this section focuses on the web view, with mobile views omitted as they closely resemble the student mobile view.
- The Admin Panel is accessible only through the web, and all screenshots will reflect the web interface.

Each panel will be broken down by feature, with screenshots attached where applicable, for either mobile or web.

I. Student Panel

Note: The web view is generally similar to the teacher's web view.

- **Login Page:** The login page allows users to access the platform by selecting their user type: Student or Teacher. If student then they will enter their matricule and password but if teacher they will enter their email and password.
- **Course Lists Page:** Shows the list of all the courses the student is enrolled in. Students can also join a course using the join course modal.

[Placeholder for Login Page Image, Course Lists Page Image, and Join Course Modal Image]

- **Course Details Page:** Displays course details of each course such as the title, name, creation

date etc. It is structured into 4 main tabs:

- **Announcements:** Specific to the course.
- **Content:** Where downloadable course files are stored.
- **Assignments:** Students can view, submit, and comment on assignments.
- **People:** Displays everyone in the course.

[Placeholder for Course Details Page Image] [Placeholder for Announcements Tab Image]

[Placeholder for Content Tab Image]

[Placeholder for Assignments Tab and Assignment Details Image] [Placeholder for People Tab Image]

- **General Announcements Page:** Lists public announcements (platform-wide). Students can comment, edit, or delete their comments, fostering interaction and clarification. Commenting on announcements encourages engagement and allows students to ask follow-up questions or share concerns.
- **General Announcement Details:** Shows detailed announcement content, including comments and attached files.
- **Profile Page:** Displays student details (name, degree program) with editable fields (level, email) and non-editable fields (name, matricule).

[Placeholder for General Announcements Page Image] [Placeholder for General Announcement Details Image] [Placeholder for Profile Page Image]

- **Upcoming Deadlines:** Lists upcoming assignment deadlines for enrolled courses.
- **Notifications:**
 - Displays notifications for new content, assignments, or announcements.
- **Curriculums:** Offers a department-wise overview of the academic structure, giving students a clear glimpse into the subjects and progression they will follow throughout their time in FET.
- **About FET:** Provides FET's history, departments, and faculty details.

[Placeholder for Upcoming Deadlines Image] [Placeholder for Notifications Image] [Placeholder for Curriculums Image] [Placeholder for About FET Image]

- *Note:* Web view is very similar to the Teacher Panel.

II. Teacher Panel

Note: Mobile view is similar to the student panel, so repeated visuals are not shown.

- **Course Lists Page:** Shows managed courses. Teachers can add courses via a modal. *See fig*

#for course lists view.

- *[placeholder for add course modal]*
- **Course Details Page:** ○ Same structure as student view (Announcements, Content, Assignments and student assignment submissions, People), but with extended functionality to create or upload materials.
 - *[Placeholder for Course Details Popup Modal Image] [Placeholder for Creation Pages Image] [Placeholder for Select Creation Type Modal Image] [Placeholder for Assignment Details Page Image]*
- **General Announcements Page:** *Same as student panel view, see student panel view.*
- **General Announcement Details:** *Same as student panel view, see student panel view.*
- **Profile Page:** Displays teacher details such as name, email, and other professional info.
 - *[Placeholder for Profile Page Image]*
- **Upcoming Deadlines:** Lists upcoming deadlines for all of a teacher's managed courses.
 - *[Placeholder for Upcoming Deadlines Image]*
- **Notifications:** Shows course-related notifications such as uploading new content or creating announcements.
 - *[Placeholder for Notifications Image]*
- **Curriculums:** *same as student panel*
- **About FET:** *same as student panel*

III. Admin Panel (Web View Only)

- **Manage Announcements:** Admins can view, create, edit, or delete announcements.
 - *[Placeholder for Manage Announcements Page Image including announcement creation page]*
- **Manage Students:** Lists students with search (by matricule) and filter (by level) options. Admins can add, edit, or delete students. He can either add a single student or add multiple students via csv or excel file upload.
 - *[Placeholder for Students List Page Image] [Placeholder for Add Student Modal Image] [Placeholder for Add Single Student Modal Image] [Placeholder for Add Students via CSV Modal Image]*
- **Manage Teachers:** Lists teachers with search and filter options. Admins can add, edit, or delete teachers. He can either add a single teacher or add multiple teachers via csv or excel file upload.
 - *[Placeholder for teachers List Page Image] [Placeholder for Add teachers Modal*

Image] [Placeholder for Add Single teachers Modal Image] [Placeholder for Add teachers via CSV Modal Image]

- **Manage Admins:** Displays a list of admins in the system. Super admins can add, edit, or delete other admins.
 - [Placeholder for Manage Admins Page Image]

4.5. Evaluation of the Solution and Comparative Analysis

To evaluate the effectiveness of the FET SPACE system, a comparative analysis was conducted against both tools previously used in FET (such as Google Classroom and WhatsApp) and a broader set of established school management and learning management systems (SMS/LMS). These included platforms such as PowerSchool, Blackbaud Education, Alma SIS, Fedena, Classe365, EduPage, Canvas, Google Classroom, and Microsoft Teams for Education.

4.5.1. Comparative Analysis of Existing Systems

The goal of this analysis was to identify how existing platforms align with or fall short of FET's university-level academic and operational requirements.

Key findings include:

- **PowerSchool** and **Blackbaud** offer comprehensive SIS and LMS features but are **expensive**, costing between \$10,000 and \$100,000 annually—well beyond the budget of FET.
- **Alma SIS** and **EduPage** are more affordable but **focus on K–12 education**, with simplified academic structures and heavy emphasis on parental oversight, which do not fit FET's needs.
- **Fedena** and **Classe365** provide low-cost solutions with basic SIS-LMS integration but **lack critical features** such as access to **past and practice questions** and customizable school-wide announcements.
- Platforms like **Canvas**, **Google Classroom**, and **Microsoft Teams** are excellent in course material sharing and quizzes but **lack support for general announcements** and **do not include student information management (SIM)** or access to historical exam content.

Table 2 summarized feature support across these systems, with  and  indicating presence or absence of key functionalities (e.g., course announcements, SIM, analytics, past questions, etc.).

Table 2: Summarized comparison of the school management system FET space and the existing solutions

Platform	SIM	Course Management	Timetable	Attendance	Gradebook	Communication	Financial	Course Announcements	General Announcements	Past Questions	Practice Questions
Power School	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗
Blackboard Education	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗
Alma SIS	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗
Fedena	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗
Classe 365	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗
EduPage	✓	✓	✓	✓	✓	✓	✗	✓	✗	✗	✗
Canvas (Instructure)	✗	✓	✓	✓	✓	✓	✗	✓	✗	✗	✓
Google Classroom	✗	✓	✓	✓	✓	✓	✗	✓	✗	✗	✓
Microsoft Teams for Education	✗	✓	✓	✓	✓	✓	✗	✓	✗	✗	✓
FET Space	✓	✓	✓	✓	✓	✓	✗	✓	✓	✓	✓

4.5.2. FET SPACE as a Contextual Solution

In contrast to the platforms above, **FET SPACE** was developed specifically for the Faculty of Engineering and Technology and designed to reflect its academic workflow, departmental structure, and common student challenges.

Key strengths include:

- Centralized academic communication through both school-wide and course-level announcements

- Access to past and practice questions to support exam readiness
- Role-based access and views (student, teacher, admin) within a unified system
- Offline functionality to accommodate students with limited internet access
- Integration with the University of Buea's academic database, ensuring accurate and structured course and user data

The system offers the **right balance of LMS and SIS features**, eliminating the need for fragmented tools like WhatsApp groups (which lack structure and traceability) and bridging the gaps left by platforms like Google Classroom (which lacks general announcements and offline support).

These results validate FET SPACE as a robust, cost-effective, and context-specific solution for university-level academic management.

While existing platforms offer robust features, they often fall short in affordability, academic alignment, or contextual relevance to FET. FET SPACE fills this gap with a tailored, role-based academic management system that merges core LMS and SIS functionalities with the unique needs of engineering education in a developing context.

Future development may focus on integrating financial management modules and further analytics to match the depth of high-end systems like PowerSchool—while maintaining cost efficiency and local adaptability.

4.6 Deployment and Execution Instructions

The deployment of the FET SPACE system was structured to support scalability, modularity, and ease of access across its three core components: the web frontend, mobile application, and backend server, all connected to a centralized PostgreSQL database.

Web Frontend (React + Vite)

The web platform, used by both **students and teachers**, was developed using **React** and built with **Vite** for optimized performance. It is hosted on a **cloud-based static hosting service** such as **Vercel**. Students and teachers access the system via the **same URL**, with views dynamically toggled based on the user's role after authentication.

Mobile Application (React Native + Expo)

The mobile app, primarily intended for students, was built with **React Native** using the **Expo**

framework for efficient cross-platform development. The final app is packaged and deployed to mobile platforms such as the **Google Play Store** and **Apple App Store**, providing accessible and responsive mobile access.

Backend Server (Node.js + Express + TypeScript)

The backend API was developed using **Node.js** and **Express**, with **TypeScript** ensuring type safety. It is hosted on a cloud platform such as **Heroku** or a VPS and is responsible for managing user authentication, data access, and all server-side logic. The API communicates securely with the frontend and mobile clients.

Database (PostgreSQL with Prisma)

A **PostgreSQL** database stores structured data such as user records, course details, and announcements. The database is hosted on supabase through their managed cloud service and is accessed using **Prisma ORM** for seamless integration with the backend and simplified schema management.

Access and Routing

- **Students and Teachers** access the system via a **shared application URL** (e.g., <https://fetspace.com>), with views automatically adjusted based on role.
- **Admins** have a **dedicated portal** (e.g., <https://admin.fetspace.com>) for managing users, content, and system-wide configurations.

4.7. Partial Conclusion

The implementation of the FET SPACE School Management System successfully delivers a functional, user-friendly platform that meets stakeholder needs. The results demonstrate its potential to revolutionize educational management at FET, with further refinements planned for future iterations.

CHAPTER 5. CONCLUSION AND FURTHER WORKS

5.1. Summary of Findings

The FET SPACE School Management System project has successfully developed a centralized, user-friendly platform tailored to the specific needs of the Faculty of Engineering and Technology (FET) within an educational institution. As detailed in Chapter 4, the system integrates essential functionalities such as course management, announcement dissemination, assignment handling, and user profile management across both mobile and web interfaces. The implementation aligns with the objectives outlined in Chapter 1, addressing the fragmentation and inefficiency of existing tools. Stakeholder feedback and system testing have confirmed that the platform meets core requirements, including real-time announcements and streamlined administrative processes, thereby enhancing the educational experience at FET.

5.2. Contribution to Engineering and Technology

The FET SPACE system makes a significant contribution to engineering and technology by providing a tailored solution that addresses the specific operational and communication needs of educational institutions. For schools, this means improved administrative efficiency, streamlined teacher-student interaction, and a centralized platform that supports academic management in a way that general-purpose tools like Google Classroom or WhatsApp cannot. FET SPACE includes features such as role-based access, internal messaging, and data centralization—all essential for running an institution effectively. Compared to WhatsApp, it offers structured, secure, and scalable data management, making it far more suitable for academic environments.

From a broader perspective, the system serves as a model for custom-built school management platforms, demonstrating how software can be designed to meet the unique challenges faced by institutions—especially in resource-constrained settings. It encourages the adoption of technology in environments that might otherwise rely on informal or fragmented systems. The development and implementation of FET SPACE also offer valuable insights to others aiming to build similar systems, showing the importance of user-specific design, accessibility, and sustainability. Additionally, it contributes to engineering education by providing a real-world example of software development that bridges technical innovation with practical institutional needs.

Ultimately, FET SPACE is not just a tool—it's a framework for modernizing educational management, and it opens up pathways for further innovation in how schools and colleges operate.

5.3. Recommendations

To ensure the widespread adoption and optimal performance of the FET SPACE system, the following recommendations are proposed:

- i. **Mandatory Usage by FET:** The faculty should mandate the use of the system among teachers, since students will only engage if their teachers are using it. Clear direction from leadership will help ensure consistent use across the institution.
- ii. **Training Programs:** FET should conduct training sessions for students and staff to improve their proficiency with the platform. Research and literature reviews highlight that usability difficulties are a common reason for staff resistance, and targeted training can address this challenge effectively.

5.4. Difficulties Encountered

The development of the FET SPACE system encountered several obstacles:

- i. **Light/Power Outages:** Frequent power outages sometimes slowed down development as both the network connectivity and battery power depended on electricity. This was mitigated by consistently saving work locally and utilizing GitHub for version control, pushing code to GitHub after each significant update to ensure data preservation and easy recovery.
- ii. **Network Connection Problems:** Unstable internet connectivity impeded testing and deployment activities. The team addressed this by scheduling development during periods of stable network availability and employing local testing environments with Thunder Client, complemented by offline coding practices.
- iii. **Managing Diverse User Functionalities and System Flow within Time Constraints:** A particularly challenging aspect was developing and integrating functionalities for the system's three distinct user roles: students, teachers, and administrators. It proved difficult and personally demanding to adequately design and implement the unique features for each user type, especially as I was also striving to paint a complete picture of the system's overall flow and clearly outline what each individual user would do within that flow, all while adhering to the specified project timeline.

5.5. Further Works

To enhance the FET SPACE system and extend its capabilities, the following areas are recommended

for future development:

- i. **Attendance Tracking:** Introduce features for marking attendance and generating attendance reports, enabling teachers and admins to monitor student participation effectively.
- ii. **Timetable Generation:** Develop a module allowing admins to create timetables by inputting halls and courses, with personalized alerts for users based on their enrolled courses to improve scheduling awareness.
- iii. **Lecturer Rating:** Implement a feature for students to rate lecturers for specific courses, facilitating constructive feedback and supporting continuous improvement in teaching quality.
- iv. **Blog/News Feed Module:** Introduce a built-in blog or news section where administrators or staff can publish summaries of key events, meetings, or announcements (e.g., FET Open Day, faculty board meetings). This will serve as an archive for institutional memory, offering accessible updates for students and staff who missed the events or wish to review them later. It also promotes transparency and engagement within the faculty.
- v. **Scalability and Performance Improvements:** As user demand grows, the system must be made more scalable. This can be achieved through database sharding to distribute data across multiple servers, horizontal scaling by adding more nodes to handle increased load, and vertical scaling by upgrading existing server resources. Performance can be further optimized by implementing caching mechanisms, optimizing database queries, and refining the frontend to reduce load times, ensuring the system remains efficient and responsive.
- vi. **Expansion of User Roles to Include Non-Teaching Staff:** Currently, the platform is primarily tailored for teaching staff, students, and administrators. Future development should focus on integrating non-teaching staff within FET as distinct user groups, providing them with relevant functionalities. This would broaden the system's utility across the entire faculty, allowing for tailored access and features pertinent to their specific roles and contributions to the FET community.

5.6. Partial Conclusion

This chapter concludes that the FET SPACE School Management System marks a significant advancement in educational management at FET, with tangible contributions to engineering and technology. Addressing the challenges encountered and pursuing the outlined further works will strengthen its impact and sustainability within the institution.

REFERENCES

- SchoolCues. (2023, April 19). *Difference between a School Management System and Learning Management System*. Retrieved April 22, 2025, from <https://www.schoolcues.com/blog/difference-between-a-school-management-system-and-learning-management-system/>
- Bolden, R. (2011). Distributed leadership in organizations: A review of theory and research. *International Journal of Management Reviews*, 13(3), 251–269.
- Buthelezi, A. B., & Ajani, O. A. (2023). Transforming school management system using participative management approach in South Africa. *International Journal of Management in Education*. (Assumed publication; full citation unavailable.)
- Caratiquit, K. D. (2021). Web-based School Information and Publication System: A Developmental Study. *SSRN Electronic Journal*. Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3909516
- Chen, P. C., Lan, T. S., Lan, Y. H., & Hsu, H. Y. (2014). Dynamic effect of knowledge management system on school management. *Journal of Theoretical and Applied Information Technology*, 61(2), 249–253.
- Gehlawat, M. (2014). School Management Information System: An Effective Tool for Augmenting the School Practices. *New Frontiers in Education*, 47(2), 57–64. Available at: <https://www.researchgate.net/publication/315380267>
- Grepon, B. S., Baran, N. T., Gumonan, K. M. V., Martinez, A. L., & Lacs, M. L. E. (2022). Designing and implementing e-school systems: An information systems approach to school management of a community college in Northern Mindanao, Philippines. *International Journal of Computing Sciences Research*. doi: 10.25147/ijcsr.2017.001.1.74. Available at: <https://ssrn.com/abstract=4145062>
- Gurr, D. (2000). How Information and Communication Technology is Changing the Work of Principals. *International Congress of School Effectiveness and Improvement, Hong Kong*.
- O'Brien, J. A. (1999). *Management Information Systems: Managing Information Technology in the Interneted Enterprise*. McGraw-Hill.
- Pavlović, M. I. J., Randić, S., & Paunović, L. (2014). Information technologies in the contemporary school management system. *Emerald Emerging Markets Case Studies*, 4(6). doi: 10.1108/EEMCS-11-2012-0195

- Polat, M., & Arabaci, I. B. (2013). Evaluation of E-school applications as a management information system. *Elementary Education Online*, 12(2), 320–333. doi: 10.17051/ieo.49201
- PricewaterhouseCoopers (PwC). (2004). *Transforming Schools with ICT: The Report to DfES*. PricewaterhouseCoopers LLP.
- Shah, M. (2014). Impact of Management Information Systems (MIS) on School Administration: What the Literature Says. *Procedia - Social and Behavioral Sciences*, 116, 2799–2804. Available at: ScienceDirect.
- Telem, M., & Buvitski, T. (1995). The potential impact of information technology on the high school principal's role: A preliminary investigation. *Studies in Educational Administration*, 63, 23–37.
- Telem, M. (1999). A case study of the impact of school administration computerization on the department head's role. *Journal of Research on Computing in Education*, 31(4), 385–401.
- Visscher, A. J. (1996b). Information technology in educational management as an emerging discipline. *International Journal of Educational Research*, 25(4), 291–296.
- JayaSudha, A. R., & Arunachalam, C. P. (2023). School Management System. *Journal of Advance Research in Mobile Computing*, 5(1). HBRP Publication. Corresponding Author E-Mail: sudhahindusthan.backup@gmail.com.
- Writer, G. (2021, September 16). What is an e-School management system? *GoSchooler*. <https://bit.ly/3RtXZM1>.
- Shah, M. (2013). Impact of management information systems (MIS) on school administration: What the literature says. *5th World Conference on Educational Sciences - WCES 2013*. Elsevier Ltd. Open access under CC BY-NC-ND license.
- Balcita, J., & Palaoag, T. (2020). [Details inferred from context; full citation not provided in prompt].
- Breiter, A., & Light, D. (2006). [Details inferred from context; full citation not provided in prompt].
- Reddy, P., & Rathna, G. (2020). [Details inferred from context; full citation not provided in prompt].
- Durnali, M. (2013). [Details inferred from context; full citation not provided in prompt].
- Yıkıcı, B., et al. (2019). [Details inferred from context; full citation not provided in prompt].
- Grepon, B. G. S., et al. (2021). Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines. Received: February 19, 2021; Accepted: May 29, 2021.